

CA-112: Programming Using C++

1. DEEP CORE NOTES

Fundamentals

- C++: Object-oriented extension of C by Bjarne Stroustrup (1979). Supports procedural + OOP + generic programming.
- Token: Smallest unit — keywords, identifiers, literals, operators, punctuators.
- Identifier rules: Start with letter/underscore, no special chars, case-sensitive.
- Data types: int(2/4B), float(4B), double(8B), char(1B), bool(1B), void.
- Scope resolution :: — accesses global var when local var has same name.
- Reference variable: Alias for existing var. `int &r = a;` `r` changes `a`.
- Default arguments: `void f(int x=10);` — must be rightmost params.
- Inline function: Substituted at call point. Reduces overhead. `inline returnType funcName(){}.`
- Function overloading: Same name, different params (type/count). Resolved at compile-time.
- new/delete: Dynamic memory. `int *p=new int(5); delete p; int *a=new int[n]; delete[] a;`

OOP Pillars

- Encapsulation: Data + functions bundled in class. Access via public/private/protected.
- Abstraction: Hiding implementation, showing interface. Achieved via access specifiers.
- Inheritance: Derived class inherits base class. Types: single, multiple, multilevel, hierarchical, hybrid.
- Polymorphism: Many forms. Compile-time (overloading) + Run-time (virtual functions).

Class & Object

- Class: Blueprint. Object: Instance. `class A{ int x; public: void show(); }; A obj;`
- Constructor: Same name as class, no return type. Auto-called on object creation.
- Types of constructor: Default, Parameterized, Copy. `A obj2(obj1)` — copy constructor.
- Destructor: `~ClassName()`. Auto-called at object end. No params, no return, only one per class.
- Static member: Shared across all objects. Defined outside class with `ClassName::var=val;`
- Friend function: Not a member but has access to private/protected. `friend void f(A);`
- `this` pointer: Points to current object. Used to resolve name conflicts.

Inheritance Syntax

```
class Derived : access_specifier Base { ... };
```

- Access in inheritance: public→public, protected→protected, private→inaccessible (for public inheritance)
- Virtual base class: Used in diamond problem. `virtual` keyword prevents duplicate copy.

Polymorphism & Virtual

- Virtual function: Declared with `virtual` in base. Overridden in derived. Enables run-time polymorphism.
- Pure virtual function: `virtual void f()=0;` Makes class abstract (cannot instantiate).
- vtable: Compiler creates table of virtual function pointers for each class.

Operator Overloading

- Redefine operators for user-defined types. `returnType operator+(A);` or as friend.
- Cannot overload: `::` `.` `*` `?:` `sizeof`

Templates

- Function template: `template<class T> T max(T a, T b){return a>b?a:b;}`
- Class template: `template<class T> class Stack{T arr[100]; int top;};`

Exception Handling

- `try{...}catch(type e){...}throw expr;` — Separates error from logic.
- `catch(...)` — catches all exceptions.

File I/O

- `ofstream`: write. `ifstream`: read. `fstream`: both. `#include<fstream>`

```
ofstream f("a.txt"); f<<"data"; f.close();  
ifstream f("a.txt"); string s; getline(f,s);
```
- File modes: `ios::in`, `ios::out`, `ios::app`, `ios::binary`, `ios::ate`

STL

- Containers: vector, list, deque, set, map, stack, queue, priority_queue
- Iterators: Input, Output, Forward, Bidirectional, Random.

- Algorithms: sort(), find(), reverse(), count() — #include<algorithm>

Key Tables

Concept	Keyword	Key Feature
Inline fn	inline	No call overhead
Virtual fn	virtual	Run-time poly
Pure virtual	=0	Abstract class
Friend	friend	Access private
Template	template<class T>	Generic code

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. What is the difference between C and C++? (2M)

- C: Procedural. C++: OOP + procedural. C++ adds classes, objects, inheritance, polymorphism, exception handling, templates, references. C uses malloc/free; C++ uses new/delete.

Q2. What is reference variable? Give example. (2M)

- An alias for existing variable. `int a=10; int &r=a; r=20; // a becomes 20.` No separate memory. Must be initialized at declaration.

Q3. Explain default arguments with example. (3M)

- Allows functions to be called without some args. `void show(int x, int y=10){ cout<<x<<y; } show(5) → x=5,y=10.` Rules: Default args must be rightmost. Cannot redefine in definition if declared in prototype.

Q4. What is inline function? Advantages and limitations. (3M)

- Inline function: compiler replaces call with function body. `inline int sq(int x){return x*x;}` Advantages: No call overhead, faster. Limitations: Code size increases, not suitable for loops/recursion/large functions.

Q5. Explain function overloading with example. (3M)

- Same function name, different param types or count. `int add(int,int); float add(float,float);` Resolved at compile-time (static binding). Return type alone NOT sufficient to overload.

Q6. What is constructor? Types of constructors? (5M)

- Constructor: Special function, same name as class, no return type. Auto-invoked on object creation.
- 1. Default: `A(){}` — no params. 2. Parameterized: `A(int x){}` — initializes with values. 3. Copy: `A(A &obj){}` — copies another object.

Q7. What is destructor? (2M)

- `~ClassName()`. No params, no return type. Only one per class. Auto-called when object goes out of scope. Used to free dynamic memory.

Q8. Explain this pointer. (3M)

- Implicit pointer available inside member functions pointing to current object. Used to resolve name conflict between member and param. `return *this` — enables method chaining.

Q9. What is friend function? (3M)

- Not a member of class but has access to private/protected members. Declared inside class with friend keyword. `friend void show(A obj);` Defined outside without `::`. Used for operator overloading involving two different classes.

Q10. Explain single and multiple inheritance with syntax. (5M)

- Single: `class B:public A{}` — one base. Multiple: `class C:public A, public B{}` — two bases. Multiple inheritance may cause ambiguity → resolved using virtual base class or scope resolution.

Q11. What is virtual function? Why needed? (5M)

- Declared in base with virtual. Overridden in derived. Base pointer to derived object calls derived version (run-time polymorphism). Without virtual: base pointer calls base version always (static binding). vtable stores function pointers.

Q12. What is abstract class? Pure virtual function? (3M)

- Pure virtual: `virtual void f()=0;` Class with ≥1 pure virtual = abstract. Cannot instantiate abstract class. Forces derived to implement all pure virtual functions.

Q13. Explain operator overloading. Rules. (5M)

- Redefining operator for user-defined types. `returnType operator+(A);` Rules: Cannot overload `::` `.` `*` `?:` `sizeof`. Cannot change arity. Cannot create new operators. Meaning cannot change for built-in types.

Q14. What are templates? Function vs class template. (5M)

- Generic programming. Function template: `template<class T> T add(T a,T b){return a+b;}` Works for any type. Class template: `template<class T> class Stack{T arr[100];};` Instantiated as `Stack<int> s;`

Q15. Explain exception handling. try-catch-throw. (5M)

- try block: risky code. throw: raises exception. catch: handles exception. Multiple catch blocks for different types. `catch(...)` handles all. Stack unwinding: destructors called when exception propagates.

Q16. File handling in C++. Modes. (5M)

- Classes: ifstream(read), ofstream(write), fstream(both). Modes: ios::in, out, app, ate, binary. Functions: open(), close(), read(), write(), get(), put(), seekg(), seekp(), tellg(), tellp()

Q17. What is STL? Name its components. (3M)

- Standard Template Library. Components: Containers (vector, list, map, set, stack, queue), Iterators (input, output, forward, bidirectional, random), Algorithms (sort, find, reverse). Enables reusable generic code.

Q18. Static member variable and function. (3M)

- Static var: shared by all objects of class. Defined outside: int A::cnt=0; Static fn: can access only static members. Called as A::fn(); No this pointer.

Q19. Explain diamond problem. Solution? (5M)

- In hybrid inheritance: class D inherits B,C both of which inherit A. Two copies of A in D → ambiguity. Solution: Virtual base class. class B:virtual public A{} class C:virtual public A{} — only one copy of A in D.

Q20. Write C++ program to overload + operator for adding two complex numbers. (8M)

```
#include<iostream>
using namespace std;
class Complex{
    float r,i;
public:
    Complex(float a=0,float b=0){r=a;i=b;}
    Complex operator+(Complex c){
        return Complex(r+c.r, i+c.i);
    }
    void show(){cout<<r<<"+"<<i<<"i\n";}
};
int main(){
    Complex c1(3,4),c2(1,2),c3;
    c3=c1+c2;
    c3.show(); // 4+6i
}
```

3. PRACTICAL / PROGRAMS**P1: Virtual Function Demo**

```
#include<iostream>
using namespace std;
class Animal{
public:
    virtual void sound(){cout<<"Animal sound\n";}
};
class Dog:public Animal{
public:
    void sound(){cout<<"Woof\n";}
};
int main(){
    Animal *p=new Dog();
    p->sound(); // Woof (run-time poly)
    delete p;
}
```

P2: Template Function

```
template<class T>
T maxVal(T a, T b){return a>b?a:b;}
int main(){
    cout<<maxVal(3,5)<<" "<<maxVal(3.5,2.1);
}
```

P3: File Write/Read

```
#include<iostream>
#include<fstream>
using namespace std;
int main(){
    ofstream o("data.txt");
    o<<"Hello";
    o.close();
    ifstream i("data.txt");
}
```

```
string s; getline(i,s);  
cout<<s;  
i.close();  
}
```

4. LAST DAY REVISION BLOCK

- C++ = C + OOP. 4 pillars: Encapsulation, Abstraction, Inheritance, Polymorphism
- inline: no overhead | default args: rightmost | overload: compile-time | virtual: run-time
- Constructor types: Default, Parameterized, Copy | Destructor: ~Name(), one per class
- Friend: accesses private, not a member | this: current object pointer
- Templates: Generic code | try-catch-throw: exception handling
- Diamond problem → virtual base class | vtable → stores virtual fn pointers
- STL: Containers + Iterators + Algorithms

ENG-113: English for All - Oral Communication

1. DEEP CORE NOTES

Communication Basics

- Communication: Process of exchanging information, ideas, feelings between sender and receiver.
- Oral Communication: Spoken exchange. Fast, immediate feedback, personal, flexible.
- Elements: Sender → Message → Channel → Receiver → Feedback → Noise/Barrier
- Types: Verbal (oral/written), Non-verbal (body language, gestures, posture, eye contact)

Barriers to Communication

Type	Examples
Physical	Noise, distance, poor equipment
Semantic	Jargon, ambiguous words
Psychological	Bias, fear, emotions
Cultural	Customs, language differences
Organizational	Information overload, poor hierarchy

Listening Skills

- Active listening: Full attention, no interruption, feedback, summarizing.
- Types: Passive (one-way), Active (interactive), Critical (evaluation), Empathetic (emotions)
- HURIER model: Hearing, Understanding, Remembering, Interpreting, Evaluating, Responding

Speaking Skills

- Clarity, Confidence, Fluency, Correct pronunciation, Appropriate pace.
- Articulation: Clear formation of sounds. Pitch: High/Low frequency. Tone: Attitude in voice.
- Volume: Loudness adjusted for audience. Pause: Strategic silence for emphasis.

Group Discussion (GD)

- GD tests: Communication, leadership, knowledge, teamwork, analytical thinking.
- Initiate: State problem, give direction. Conclude: Summarize all views logically.
- Do: Stay relevant, be assertive not aggressive, listen to others, use facts.
- Don't: Interrupt rudely, deviate from topic, dominate, use wrong grammar.

Interview Skills

- Types: One-to-one, Panel, Telephonic, Video, Stress, Structured, Unstructured, Behavioural
- Before: Research company, prepare FAQs, dress code, documents ready.
- During: Firm handshake, eye contact, listen carefully, structured answers (STAR method).
- STAR: Situation, Task, Action, Result — for behavioral questions.
- Common Qs: Tell me about yourself, strengths/weaknesses, why this company, where do you see yourself.

Presentation Skills

- Structure: Introduction (hook+agenda), Body (key points), Conclusion (summary+CTA).
- 7-38-55 Rule: 7% words, 38% tone, 55% body language (Mehrabian).
- Visual aids: Reinforce content, not replace. Keep slides simple, readable.
- Handling questions: Listen fully, pause, answer precisely, say 'I'll find out' if unsure.

Phonetics

- Phoneme: Smallest sound unit. Vowels: a,e,i,o,u. Consonants: remaining 21.
- Syllable: Unit of pronunciation. Stress: emphasis on syllable. Primary (ˈ) Secondary (ˌ)
- Common mispronounced: often (/ˈɔːfən/), comfortable (/ˈkʌmfətl̩/), receipt (/rɪˈsiːt/)

Body Language

- Posture: Straight = confidence. Slouch = disinterest.
- Eye contact: Confidence, trust. Avoid: Staring or looking away constantly.
- Gestures: Open palms = honesty. Crossed arms = defensive.
- Proxemics: Space study. Intimate(<0.5m), Personal(0.5-1.2m), Social(1.2-3.6m), Public(>3.6m)

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. Define communication. List its elements. (2M)

- Process of transmitting and receiving information between sender and receiver. Elements: Sender, Message, Encoding, Channel, Decoding, Receiver, Feedback, Noise.

Q2. What is oral communication? Advantages. (3M)

- Face-to-face or voice-based exchange. Advantages: Immediate feedback, personal, flexible, natural, builds rapport. Disadvantage: No permanent record, easily distorted.

Q3. What are barriers to communication? (5M)

- Physical: noise, distance. Semantic: jargon, ambiguity. Psychological: stress, emotions, bias. Cultural: customs, language. Organizational: overload, poor structure. Solution: Clear language, feedback, empathy, appropriate channel.

Q4. Explain active vs passive listening. (3M)

- Active: Full attention, ask questions, summarize, feedback. Passive: One-way, no response. Active = better understanding, reduces errors, builds trust.

Q5. What is non-verbal communication? (3M)

- Conveying message without words. Includes: body language, gestures, facial expressions, posture, eye contact, space (proxemics), tone of voice. 55% of meaning from non-verbal (Mehrabian).

Q6. Describe Group Discussion. What qualities are tested? (5M)

- GD = group of 8-12 discussing topic for 15-20 min. Tests: Communication skills, leadership, knowledge, teamwork, listening, confidence. Roles: Initiator, Contributor, Summarizer. Key: Be assertive, not aggressive.

Q7. How to give an effective presentation? (5M)

- Structure: Intro (hook+objectives), Body (3 main points), Conclusion (summary+CTA). Use: eye contact, clear voice, visuals. Avoid: reading slides, filler words (um,uh), monotone.

Q8. Types of interviews. (3M)

- One-to-one, Panel, Telephonic, Video, Stress, Structured (fixed Qs), Unstructured (open-ended), Behavioral (past experience), Exit.

Q9. What is STAR method in interviews? (3M)

- Situation: Context/background. Task: Your responsibility. Action: Steps you took. Result: Outcome. Used for behavioral Qs like 'Tell me about a time when...' Gives structured, evidence-based answers.

Q10. What is phonetics? Types of sounds? (3M)

- Study of speech sounds. Phoneme = smallest unit. Vowel sounds (20 in English), Consonant sounds (24). IPA = International Phonetic Alphabet. Stress marks: primary (') secondary (,).

Q11. Explain 7-38-55 rule of communication. (2M)

- Mehrabian's rule: 7% = words (content), 38% = voice tone (paralanguage), 55% = body language. Most communication is non-verbal. Applies in face-to-face only.

Q12. What is proxemics? (2M)

- Study of personal space in communication. Intimate zone <0.5m, Personal 0.5-1.2m, Social 1.2-3.6m, Public >3.6m (Hall's zones).

Q13. How to handle stage fear? (3M)

- Preparation: Know content well. Practice: Rehearse aloud. Breathing: Deep breaths calm nerves. Visualization: Imagine success. Start slow: First sentence sets tone. Focus on message, not audience reaction.

Q14. What is feedback in communication? (2M)

- Receiver's response to sender's message. Makes communication two-way. Types: Verbal, Non-verbal, Written. Positive feedback reinforces; corrective feedback improves.

Q15. Difference between formal and informal communication. (3M)

- Formal: Official channel, structured, documented. Eg: meetings, presentations. Informal (grapevine): Unofficial, fast, distorted. Formal flows: Downward, Upward, Horizontal.

Q16. What is paralanguage? (2M)

- Vocal but non-verbal aspect of speech: pitch, tone, volume, rate, pauses, emphasis, quality of voice. Conveys emotion, attitude. Same words different tone = different meaning.

Q17. How to introduce yourself professionally? (3M)

- Name → Educational background → Professional experience → Skills → Interests → Goal. Keep under 90 seconds. Confident, clear, no memorized script. End with 'I'm excited about this opportunity'.

Q18. What is kinesics? (2M)

- Study of body movement in communication. Includes: gestures, facial expressions, posture, eye contact, head movements. Emblems: direct meaning (thumbs up). Illustrators: support speech.

Q19. Difference between hearing and listening. (2M)

- Hearing: Passive, physical process, receiving sound waves. Listening: Active, mental process, interpreting meaning. Hearing is involuntary; listening is intentional and requires concentration.

Q20. Tips for effective GD participation. (5M)

- Start with data/fact/question. Maintain eye contact with all. Build on others' points. Use connector phrases: 'I agree because...', 'Adding to that...'. Summarize at end. Don't interrupt; wait for pause. Speak 3-4 times minimum.

4. LAST DAY REVISION BLOCK

- Communication: Sender→Message→Channel→Receiver→Feedback
- Barriers: Physical, Semantic, Psychological, Cultural, Organizational
- 7-38-55: words(7%), tone(38%), body language(55%) — Mehrabian
- GD: Initiate, Contribute, Listen, Conclude. Be assertive not aggressive.
- STAR: Situation, Task, Action, Result
- Proxemics: Intimate<0.5m | Personal 0.5-1.2m | Social 1.2-3.6m | Public>3.6m
- Active listening > Passive | Kinesics = body movement | Proxemics = space

ES-115: पर्यावरण अभ्यास (Environmental Studies - Marathi)

1. सखोल मुख्य नोट्स

पर्यावरण - मूलभूत संकल्पना

- पर्यावरण: सजीव व निर्जीव घटकांचे परस्परावलंबी जाळे. Biotic + Abiotic घटक.
- परिसंस्था (Ecosystem): जिवसृष्टी + भौतिक पर्यावरण यांचे एकत्रित कार्यात्मक एकक.
- जैवविविधता: पृथ्वीवरील सजीव प्रजातींची विविधता. Genetic, Species, Ecosystem तीन पातळ्या.
- अन्नसाखळी: उत्पादक → प्राथमिक ग्राहक → द्वितीयक ग्राहक → तृतीयक ग्राहक → विघटक
- अन्नजाळे: अनेक अन्नसाखळ्यांचे जटिल जाळे. अधिक स्थिरता देते.

नैसर्गिक संसाधने

- नवीकरणीय: सौर ऊर्जा, वारा, जल, जैविक — पुनर्निर्माण होतात.
- अनवीकरणीय: कोळसा, खनिज तेल, नैसर्गिक वायू — मर्यादित साठा.
- वन संसाधने: मातीची धूप रोखणे, जलप्रवाह नियमन, O₂ उत्पादन, जैवविविधता.
- जल संसाधने: पाण्याचा फेरफटका (Water Cycle): Evaporation → Condensation → Precipitation → Runoff

प्रदूषण

प्रदूषण प्रकार	मुख्य कारणे	परिणाम
वायू प्रदूषण	वाहने, उद्योग, CO ₂ , SO ₂	आम्लपर्जन्य, श्वसनरोग
जल प्रदूषण	सांडपाणी, कीटकनाशके	कॉलरा, विषबाधा
भूमी प्रदूषण	प्लास्टिक, रासायनिक खते	माती निकस, पिकनाश
ध्वनी प्रदूषण	वाहने, यंत्रे, फटाके	बहिरेपणा, ताण
किरणोत्सारी	अणुकेंद्र, X-ray	कर्करोग, अनुवांशिक दोष

हवामान बदल व ग्लोबल वॉर्मिंग

- हरितगृह परिणाम: CO₂, CH₄, N₂O, H₂O वायू सूर्याची उष्णता अडवतात → तापमान वाढ.
- ग्लोबल वॉर्मिंग परिणाम: हिमनदी वितळणे, समुद्र पातळी वाढ, तीव्र हवामान, जैवविविधता नुकसान.
- पॅरिस करार (2015): 2°C पेक्षा कमी तापमान वाढ ठेवणे. 195 देश सहभागी.

जैवविविधता संरक्षण

- In-situ: नैसर्गिक अधिवासात संरक्षण. राष्ट्रीय उद्यान, अभयारण्य, जैव राखीव क्षेत्र.
- Ex-situ: नैसर्गिक अधिवासाबाहेर. प्राणीसंग्रहालय, बोटॅनिकल गार्डन, जीन बँक.
- भारतातील बायोस्फेअर रिझर्व्ह: 18. प्रमुख: नीलगिरी, नंदादेवी, सुंदरबन.

पर्यावरण कायदे व संस्था

- पर्यावरण संरक्षण कायदा 1986: मुख्य कायदा. सर्व प्रदूषण नियंत्रण.
- वायू प्रदूषण कायदा 1981, जल प्रदूषण कायदा 1974.
- CPCB: केंद्रीय प्रदूषण नियंत्रण मंडळ. MPCB: महाराष्ट्र प्रदूषण नियंत्रण मंडळ.
- वन्यजीव संरक्षण कायदा 1972: प्रजातींचे संरक्षण. IUCN रेड लिस्ट: Extinct, Endangered, Vulnerable.

सांडपाणी व घनकचरा व्यवस्थापन

- 3R तत्व: Reduce (कमी करा), Reuse (पुनर्वापर), Recycle (पुनर्प्रक्रिया).
- Vermicomposting: गांडूळ खत. Biogas: जैव वायू — CO₂ + CH₄ पासून ऊर्जा.
- E-waste: इलेक्ट्रॉनिक कचरा. Pb, Hg, Cd — विषारी धातू. अधिकृत केंद्रात निपटारा.

शाश्वत विकास

- Sustainable Development: वर्तमान गरजा पूर्ण करताना भविष्यातील पिढ्यांच्या गरजांशी तडजोड न करणे (Brundtland 1987).
- SDGs: 17 शाश्वत विकास उद्दिष्टे (UN 2015). 2030 पर्यंत पूर्ण करण्याचे लक्ष्य.

2. 20 अत्यंत महत्वाचे प्रश्न व उत्तरे

Q1. पर्यावरण म्हणजे काय? (2M)

- एखाद्या सजीवाभोवती असलेल्या सर्व जैविक (Biotic) व अजैविक (Abiotic) घटकांचे एकत्रित रूप म्हणजे पर्यावरण होय. यात हवा, पाणी, माती, इतर सजीव समाविष्ट असतात.

Q2. परिसंस्था म्हणजे काय? उदाहरण द्या. (3M)

- एका विशिष्ट भागातील सजीव समुदाय (Biotic) व त्यांचे भौतिक पर्यावरण (Abiotic) यांच्यातील परस्परावलंबी एकक म्हणजे परिसंस्था. उदाहरण: तलाव, जंगल, समुद्र, वाळवंट. घटक: उत्पादक, ग्राहक, विघटक.

Q3. अन्नसाखळी व अन्नजाळे यातील फरक. (3M)

- अन्नसाखळी: एकेरी रेषीय ऊर्जा प्रवाह. उत्पादक → तृणभक्षी → मांसभक्षी. अन्नजाळे: अनेक अन्नसाखळ्यांचे जटिल जाळे. अधिक प्रजाती = अधिक स्थिरता. वास्तवात अन्नजाळेच असते.

Q4. जैवविविधता म्हणजे काय? धोके सांगा. (5M)

- पृथ्वीवरील सर्व प्रजातींच्या अनुकीय, प्रजाती व परिसंस्था विविधतेस जैवविविधता म्हणतात. धोके: अधिवास नाश, जास्त शिकार, प्रदूषण, हवामान बदल, परदेशी प्रजातींचे आक्रमण. संरक्षण: in-situ (राखीव क्षेत्र), ex-situ (प्राणीसंग्रहालय).

Q5. हरितगृह परिणाम व ग्लोबल वॉर्मिंग स्पष्ट करा. (5M)

- CO₂, CH₄, N₂O वायू पृथ्वीच्या वातावरणात उष्णता अडवतात → हरितगृह परिणाम. यामुळे सरासरी तापमान वाढते = ग्लोबल वॉर्मिंग. परिणाम: हिमनदी वितळणे, समुद्र पातळी वाढ, अतिवृष्टी/दुष्काळ, जैवविविधता नाश.

Q6. वायू प्रदूषण कारणे व उपाय. (5M)

- कारणे: वाहने (CO, NO_x), उद्योग (SO₂), जंगलतोड, कोळसा ज्वलन. परिणाम: आम्लपर्जन्य, श्वसनरोग, ओझोन क्षरण. उपाय: सार्वजनिक वाहतूक, स्वच्छ ऊर्जा, औद्योगिक फिल्टर, वृक्षारोपण.

Q7. जल प्रदूषण थांबवण्याचे उपाय. (3M)

- सांडपाण्यावर प्रक्रिया (STP). कीटकनाशके कमी वापर. प्लास्टिक निर्मूलन. नदी स्वच्छता अभियान. कारखान्यांचे सांडपाणी थेट नदीत सोडणे बंद. पाण्याचा पुनर्वापर.

Q8. नवीकरणीय व अनवीकरणीय ऊर्जा स्रोत. (3M)

- नवीकरणीय: सौर, वारा, जल, भू-औष्णिक, जैव वायू — पुन्हा निर्माण होतात, प्रदूषण कमी. अनवीकरणीय: कोळसा, खनिज तेल, नैसर्गिक वायू — मर्यादित, प्रदूषण जास्त. भविष्य: नवीकरणीयवर अवलंबून राहणे गरजेचे.

Q9. शाश्वत विकास संकल्पना. (3M)

- वर्तमान पिढीच्या गरजा पूर्ण करताना भविष्यातील पिढ्यांच्या गरजांशी तडजोड न करणारा विकास. Brundtland आयोग 1987. 3 स्तंभ: आर्थिक, सामाजिक, पर्यावरणीय. SDGs: 17 उद्दिष्टे.

Q10. पर्यावरण संरक्षण कायदा 1986. (3M)

- भारतातील सर्वसमावेशक पर्यावरण कायदा. Bhopal Gas Tragedy नंतर लागू. केंद्र सरकारला प्रदूषण नियंत्रण अधिकार. MoEFCC द्वारे अंमलबजावणी. EIA (पर्यावरण परिणाम मूल्यांकन) अनिवार्य.

Q11. In-situ व Ex-situ संरक्षण. (3M)

- In-situ: नैसर्गिक अधिवासात संरक्षण. राष्ट्रीय उद्यान, अभयारण्य, बायोस्फेअर रिझर्व्ह. Ex-situ: बाहेर संरक्षण. प्राणीसंग्रहालय, बोटॅनिकल गार्डन, Seed Bank, DNA Bank.

Q12. ओझोन थर क्षरण. (3M)

- O₃ थर UV-B किरण शोषतो. CFC (Chlorofluorocarbon) वायू ओझोन नष्ट करतात. Antarctic Ozone Hole. मॉन्ट्रियल करार 1987: CFC बंदी. UV-B वाढल्यास: त्वचा कर्करोग, मोतीबिंदू, DNA नुकसान.

Q13. भूमी प्रदूषण व E-waste. (3M)

- भूमी प्रदूषण: प्लास्टिक, रासायनिक खते, कीटकनाशके, घनकचरा. E-waste: संगणक, मोबाईल, इलेक्ट्रॉनिक कचरा. Pb, Hg, Cd — विषारी. WEEE नियम अंतर्गत अधिकृत पुनर्वापर केंद्र.

Q14. 3R तत्व सांगा. (2M)

- Reduce: वापर कमी करा. Reuse: पुन्हा वापरा. Recycle: पुनर्प्रक्रिया करा. यास 4R (Recover) व 5R (Refuse) देखील जोडतात. कचरा व्यवस्थापनाचा मुख्य पाया.

Q15. Vermicomposting म्हणजे काय? (2M)

- गांडूळांचा वापर करून जैव कचऱ्यावर प्रक्रिया करून तयार केलेले सेंद्रिय खत म्हणजे Vermicompost (गांडूळ खत). माती सुधारते, रासायनिक खतांची गरज कमी होते.

Q16. IUCN रेड लिस्ट वर्गीकरण. (3M)

- EX: Extinct (नामशेष). EW: Extinct in Wild. CR: Critically Endangered. EN: Endangered. VU: Vulnerable. NT: Near Threatened. LC: Least Concern. DD: Data Deficient.

Q17. वन्यजीव संरक्षण कायदा 1972. (3M)

- प्राण्यांची शिकार, व्यापार, अधिवास नाश बंदी. Schedule I (सर्वोच्च संरक्षण): वाघ, सिंह, हत्ती. Project Tiger (1973), Project Elephant. TRAFFIC: वन्यजीव व्यापार नियंत्रण.

Q18. ध्वनी प्रदूषण नियंत्रण. (2M)

- मर्यादा: दिवसा 55 dB (निवासी), 75 dB (औद्योगिक). रात्री 45 dB (निवासी). उपाय: शांतता क्षेत्र, noise barriers, हॉर्न बंदी, कानाचे रक्षक. ध्वनी प्रदूषण नियम 2000.

Q19. जल प्रदूषण कायदा 1974. (2M)

- कारखान्यांचे सांडपाणी जलस्त्रोतांत सोडणे नियंत्रित. CPCB/SPCB निरीक्षण करते. ETP (Effluent Treatment Plant) अनिवार्य. उल्लंघनास दंड व तुरुंगवास.

Q20. पर्यावरण शिक्षण महत्त्व व उपाय. (5M)

- महत्त्व: जागरूकता, जबाबदारी, शाश्वत वर्तन. Stockholm Conference 1972: पर्यावरण शिक्षण ठरावाचे उद्गम. Tbilisi Declaration 1977: पर्यावरण शिक्षण तत्वे. शालेय अभ्यासक्रमात समावेश. NSS, वृक्षारोपण, स्वच्छता मोहिमा. Awareness → Attitude → Action.

4. अंतिम दिवस रिव्हिजन

- परिसंस्था = Biotic + Abiotic | अन्नसाखळी: उत्पादक → ग्राहक → विघटक
- 3R: Reduce, Reuse, Recycle | शाश्वत विकास: Brundtland 1987
- हरितगृह वायू: CO₂, CH₄, N₂O | ओझोन नाशक: CFC
- In-situ: राष्ट्रीय उद्यान | Ex-situ: प्राणीसंग्रहालय, Gene Bank
- कायदे: EPA 1986, Water 1974, Air 1981, Wildlife 1972
- CPCB: केंद्रीय | MPCB: महाराष्ट्र | IUCN: रेड लिस्ट

CA-121: Object Oriented Programming in C++

1. DEEP CORE NOTES

OOP Fundamentals

- OOP: Programming paradigm based on objects. Solves real-world problems by modeling entities as objects.
- Object: Instance of class. Has state (data members) + behavior (member functions).
- Class: User-defined data type. Blueprint for objects. Contains data members + member functions.
- Access specifiers: public (accessible anywhere), private (within class), protected (class + derived).

Core OOP Concepts

- Encapsulation: Wrapping data + functions. Data hiding via private. Accessed via public methods (getters/setters).
- Abstraction: Show only essential features. Hide complexity. Abstract class forces interface contract.
- Inheritance: 'is-a' relationship. Code reuse. Base class → Derived class.
- Polymorphism: Same interface, different behavior. Compile-time (overloading/templates), Run-time (virtual).

Constructors & Destructors (Deep)

- Constructor Initialization List: `A(int x):val(x){}` — efficient, required for const/ref members.
- Copy constructor: Called when: object passed by value, returned from function, initialized from another object.
- Shallow copy (default): Copies pointer address. Problem with dynamic memory → both point same.
- Deep copy: Allocates new memory, copies values. Must write explicit copy constructor for dynamic members.

Inheritance (Deep)

- Constructor call order: Base first, then Derived. Destructor: Derived first, then Base.
- Derived class accessing base: public members accessible directly; private NOT directly (use protected or public methods).
- Function overriding: Same signature in derived. Base pointer calls base (without virtual), derived (with virtual).

Inheritance Type	Syntax	Use
Single	<code>class D:public B</code>	One base
Multiple	<code>class D:public B1,public B2</code>	Two+ bases
Multilevel	<code>class A→B→C</code>	Chain
Hierarchical	<code>B1,B2 from A</code>	Same base
Hybrid	Combo above	Diamond issue

Polymorphism (Deep)

- Static (Compile-time): Function overloading, operator overloading, template. Resolved at compile.
- Dynamic (Run-time): Virtual function + base class pointer. Resolved at runtime via vtable.
- Pure virtual function = 0: Forces derived to implement. Abstract class cannot be instantiated.
- Override keyword (C++11): `void f() override;` — compiler error if not actually overriding.
- Final keyword (C++11): Prevents further overriding or inheritance.

Operator Overloading (Deep)

- As member: `T operator+(T);` (left operand is this). As friend: `friend T operator+(T,T);` (both explicit).
- Unary: `T operator-();` `T operator++();` (prefix) `T operator++(int);` (postfix with dummy int)
- Stream: `friend ostream& operator<<(ostream&, T);` `friend istream& operator>>(istream&, T);`

Pointer to Object

- `A *p = new A();` `p->member` (arrow); `(*p).member` (dot). `delete p;` for cleanup.
- Array of objects: `A arr[3];` or `A *arr=new A[3];` `delete[] arr;`

Type Casting in C++

- `static_cast<>`: Basic conversion. `dynamic_cast<>`: Safe downcast (polymorphic). `const_cast<>`: Remove const. `reinterpret_cast<>`: Pointer reinterpretation.

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. Difference between OOP and Procedural Programming. (3M)

- Procedural: Code organized in functions, data flows through. Top-down. C. OOP: Code organized in objects (data+functions together). Bottom-up. C++. OOP advantages: Reusability, Modularity, Security (encapsulation), Easier maintenance.

Q2. Explain encapsulation with example. (3M)

- Binding data and functions. Private data accessed via public methods. `class BankAccount{private: double bal; public: void deposit(double); double getBalance();};` User cannot directly access `bal`.

Q3. What is data abstraction? How achieved? (3M)

- Hiding internal implementation, showing only interface. Achieved via: access specifiers (private/protected), abstract classes (pure virtual), interfaces. User interacts with public methods without knowing internals.

Q4. Deep copy vs shallow copy. (5M)

- Shallow: Default copy. Copies pointer value → both objects share same memory → double delete = crash. Deep: Explicit copy constructor allocates new memory for pointer members. `class A{int *p; A(A &o){p=new int(*o.p);}};` Required when class has dynamic memory.

Q5. Constructor initialization list. When mandatory? (3M)

- Syntax: `A(int x, int y): a(x), b(y){}` Mandatory for: const data members (cannot assign after init), reference members, base class constructors, member objects without default constructor. More efficient than assignment in body.

Q6. Explain runtime polymorphism. How achieved? (5M)

- Achieved via virtual functions + base class pointer/reference. `void func(Animal *p){p->sound();}` `func(new Dog())` calls `Dog::sound()`. Mechanism: vtable (virtual function table) stores fn pointers. `vp`tr in object points to vtable. Resolved at runtime.

Q7. What is vtable and vp

tr? (3M)

- vtable: Per-class table of virtual function pointers created by compiler. `vp`tr: Per-object pointer to its class's vtable. When virtual function called via base pointer: follow `vp`tr → vtable → correct function. Overhead: one pointer per object + table per class.

Q8. Abstract class vs interface concept in C++. (3M)

- Abstract class: Has ≥1 pure virtual function. Cannot instantiate. Can have non-pure methods and data. In C++, pure abstract class (all pure virtual) = interface equivalent. Derived MUST implement all pure virtuals or remain abstract.

Q9. Explain multiple inheritance with diamond problem. (5M)

- `class A{} class B:public A{} class C:public A{} class D:public B,public C{} → D has two copies of A → ambiguity. d.A::x (which one?). Solution: virtual inheritance. class B:virtual public A{} class C:virtual public A{} → only one A in D.`

Q10. Overload ++ operator (prefix and postfix). (5M)

```
class Counter{
    int c;
public:
    Counter(int x=0):c(x){}
    Counter operator++(){++c; return *this;} // prefix
    Counter operator++(int){Counter t=*this; c++; return t;} // postfix
    int get(){return c;}
};
```

Q11. Function overriding vs overloading. (3M)

- Overloading: Same name, different params, same class, compile-time. Overriding: Same name, same params, different class (base-derived), runtime. Overriding requires virtual for runtime poly. Without virtual: static binding (base version called).

Q12. this pointer uses. (3M)

- 1. Resolve conflict: `void setX(int x){this->x=x;}` 2. Return current object: `return *this;` (chaining) 3. Pass current object to another function. const member function: this is const pointer. Cannot modify non-mutable members.

Q13. Static class members. (3M)

- Static data: One copy for all objects. Counts instances. `int A::count=0;` Static function: No this pointer. Can call without object: `A::staticFn();` Can access only static members. Used for: factory patterns, utility functions, instance counting.

Q14. Friend class. (3M)

- friend class B; inside A → B can access A's private/protected. Not symmetric (A can't access B's). Not transitive. Used for: operator overloading needing access to two classes, tightly coupled utility classes.

Q15. Explain type casting operators in C++. (3M)

- `static_cast`: int to float, upcasting. `dynamic_cast`: safe downcast in inheritance hierarchy; returns null if fails; requires polymorphic type. `const_cast`: remove/add const. `reinterpret_cast`: dangerous pointer type conversion.

Q16. new vs malloc. (3M)

- `new`: C++ operator, calls constructor, returns typed pointer, throws `bad_alloc`. `malloc`: C function, no constructor, returns `void*`, returns NULL on fail. `delete` calls destructor; `free` does not. Always use `new/delete` in C++.

Q17. What is object slicing? (2M)

- When derived object assigned to base object (not pointer): derived-specific data is 'sliced off'. Base base = derived; — loses derived members. Use pointers/references to avoid slicing: Base *p = &derived; or Base &r = derived;

Q18. Overload << operator for output. (5M)

```
class Point{
    int x,y;
public:
    Point(int a,int b):x(a),y(b){}
    friend ostream& operator<<(ostream &out, Point p){
        out<<" ("<<p.x<<" "<<p.y<<" " ";
        return out;
    }
};

int main(){Point p(3,4); cout<<p; // (3,4)
}
```

Q19. Constructors in inheritance. Order? (3M)

- Creation: Base constructor first → Derived constructor. Destruction: Derived destructor → Base destructor. Explicitly calling base: Derived(int x):Base(x){} If not called, default base constructor invoked. Virtual destructor in base ensures proper cleanup via base pointer.

Q20. Write program showing runtime polymorphism. (8M)

```
#include<iostream>
using namespace std;
class Shape{
public:
    virtual float area()=0;
    virtual void show(){cout<<"Shape: area="<<area()<<"\n";}
    virtual ~Shape(){}
};

class Circle:public Shape{
    float r;
public:
    Circle(float x):r(x){}
    float area(){return 3.14*r*r;}
};

class Rect:public Shape{
    float l,w;
public:
    Rect(float a,float b):l(a),w(b){}
    float area(){return l*w;}
};

int main(){
    Shape *s[2];
    s[0]=new Circle(5);
    s[1]=new Rect(4,6);
    for(int i=0;i<2;i++) s[i]->show();
    for(int i=0;i<2;i++) delete s[i];
}
```

4. LAST DAY REVISION BLOCK

- 4 Pillars: Encapsulation, Abstraction, Inheritance, Polymorphism
- Shallow copy: shares memory (DANGER) | Deep copy: new memory (SAFE)
- vtable: per class | vptr: per object | virtual → runtime poly
- Diamond: virtual base class | Constructor: Base first, Destructor: Derived first
- Pure virtual =0 → abstract class → cannot instantiate
- new: calls constructor, throws bad_alloc | malloc: no ctor, returns NULL
- Object slicing: use pointer/reference to avoid

CA-122: Introduction to Operating System

1. DEEP CORE NOTES

OS Fundamentals

- OS: Software that manages hardware and software resources. Interface between user and hardware.
- Functions: Process mgmt, Memory mgmt, File mgmt, Device mgmt, Security, UI.
- Types: Batch, Time-sharing, Real-time (RTOS), Distributed, Embedded, Mobile, Network.
- Kernel: Core of OS. Manages CPU, memory, devices. Types: Monolithic, Microkernel, Hybrid.
- System calls: Interface for programs to request OS services. (fork, exec, read, write, open, close)

Process Management

- Process: Program in execution. PCB (Process Control Block): stores PID, state, PC, registers, memory info.
- Process states: New → Ready → Running → Waiting → Terminated
- Process vs Thread: Process=heavy, independent memory. Thread=light, shared memory within process.
- Context switch: Saving current process state, loading next. Overhead: pure overhead (no useful work done).

CPU Scheduling

Algorithm	Preemptive?	Key Feature	Problem
FCFS	No	First come first served	Convoy effect
SJF	No	Shortest job first	Starvation
SRTF	Yes	Shortest remaining time	Starvation
Round Robin	Yes	Time quantum	High context switches
Priority	Both	Highest priority first	Starvation
MLQ	Both	Multiple queues	Complex

- Metrics: Throughput (jobs/time), Turnaround=completion-arrival, Waiting=turnaround-burst, Response=first response-arrival
- Aging: Gradually increase priority of waiting processes → prevents starvation.

Deadlock

- Deadlock: Set of processes waiting for each other's resources, none can proceed.
- Coffman conditions (all 4 must hold): 1.Mutual Exclusion 2.Hold&Wait 3.No Preemption 4.Circular Wait
- Handling: 1.Prevention (negate condition), 2.Avoidance (Banker's algo), 3.Detection & Recovery, 4.Ignore (Ostrich)
- Banker's Algorithm: OS checks safe state before allocating resources. Maintains: Available, Allocation, Max, Need matrices.

Memory Management

- Contiguous: Partitioning. Fixed (internal fragmentation), Variable (external fragmentation).
- Paging: Divide memory into fixed frames, process into pages. Page table maps page→frame. No external frag.
- Segmentation: Divide by logical units (code, data, stack). Segment table: base+limit. External frag.
- Virtual memory: Process larger than RAM. Demand paging: load pages only when needed.
- Page fault: Page not in memory → OS loads from disk. Excessive faults → Thrashing.

Page Replacement

Algorithm	Replace	Optimal?
FIFO	Oldest page	No, Belady's anomaly
LRU	Least recently used	Near optimal
Optimal/OPT	Future least used	Best but impractical
Clock	Not-recently-used bit	Approx LRU

File System

- File: Named collection of data. Attributes: name, type, size, date, permissions, location.
- Directory: Single-level, Two-level, Tree, Acyclic graph, General graph.
- File allocation: Contiguous (fast, external frag), Linked (no frag, slow), Indexed (i-node).
- UNIX: i-node = inode stores file metadata + 12 direct + single/double/triple indirect blocks.

Disk Scheduling

Algorithm	Head Movement	Key
-----------	---------------	-----

FCFS	Order of requests	Simple, high seek time
SSTF	Nearest track first	Starvation possible
SCAN	One end to other, serve	Elevator algo
C-SCAN	One direction only	Uniform wait
LOOK	To last request, reverse	SCAN variant

Process Synchronization

- Critical section: Code accessing shared resource. Race condition: Outcome depends on execution order.
- Solution requirements: 1.Mutual Exclusion 2.Progress 3.Bounded Waiting
- Semaphore: wait(s)/P: s--; if s<0 block. signal(s)/V: s++; if s<=0 wakeup.
- Mutex: Binary semaphore. 0/1 only. Only owner can unlock.
- Monitor: High-level sync. Condition variables: wait(), signal().
- Classic problems: Producer-Consumer, Readers-Writers, Dining Philosophers

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. What is OS? Functions? (3M)

- OS: Interface between user and hardware. Manages resources. Functions: Process management (scheduling), Memory management (allocation), File management (create/delete), Device management (I/O), Security/Protection, UI (CLI/GUI).

Q2. Process vs Thread. (3M)

- Process: Independent execution unit, own memory, heavy, costly context switch. Thread: Lightweight process, shares memory with parent process, fast switch, cheap creation. Multi-threading: multiple threads in one process. Benefits: Responsiveness, Resource sharing, Economy.

Q3. Process states and PCB. (5M)

- States: New (created), Ready (waiting CPU), Running (executing), Waiting/Blocked (waiting I/O), Terminated. PCB: PID, state, PC, CPU registers, memory maps, I/O status, scheduling info. Context switch: saves PCB of current, loads PCB of next.

Q4. Calculate turnaround and waiting time (FCFS example). (5M)

- FCFS: P1(0,6), P2(1,4), P3(2,2). Execute: P1(0-6), P2(6-10), P3(10-12). TAT: P1=6, P2=9, P3=10. WT: P1=0, P2=5, P3=8. Avg WT=(0+5+8)/3=4.33. Convoy effect: short jobs wait for long jobs.

Q5. Round Robin scheduling. (5M)

- Preemptive. Time quantum q. Each process gets q time then preempted. Fair, suitable for time-sharing. Problem: High context switch if q too small; becomes FCFS if q too large. Response time = $q \cdot (n-1)$ where n=processes. Optimal q usually 10-100ms.

Q6. What is deadlock? Coffman conditions? (5M)

- Deadlock: Processes waiting for each other's resources permanently. All 4 Coffman must hold: 1.Mutual Exclusion: resource non-sharable. 2.Hold & Wait: holding one, waiting another. 3.No Preemption: can't take resource away. 4.Circular Wait: $P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_n \rightarrow P_1$.

Q7. Banker's Algorithm. (5M)

- Deadlock avoidance. Before allocation, check if safe state exists. Safe state: sequence of processes that can finish one by one. Uses: Max, Allocation, Need=Max-Allocation, Available. Safety algorithm: Find process where $Need \leq Available$, simulate finish, release resources, repeat.

Q8. Paging vs Segmentation. (5M)

- Paging: Fixed size (pages/frames), physical division, no external frag, page table maps. Segmentation: Variable size (logical units), logical division, external frag, segment table (base+limit). Combined: Paged segmentation — modern OSes use this.

Q9. What is virtual memory? Page fault handling. (5M)

- Virtual memory: Allows process larger than physical RAM. Non-resident pages stay on disk. Demand paging: pages loaded only when accessed. Page fault: referenced page not in memory → OS: 1.Save state 2.Find free frame (or replace page) 3.Load page from disk 4.Update page table 5.Restart instruction.

Q10. Compare page replacement algorithms. (5M)

- FIFO: Simple, replace oldest → Belady's anomaly (more frames = more faults). LRU: Replace least recently used, efficient, needs hardware support. OPT: Replace page not used longest in future, theoretical benchmark. Clock: Approximate LRU with reference bit, practical.

Q11. What is thrashing? How to prevent? (3M)

- Thrashing: CPU spends more time paging than executing. Cause: Too many processes, insufficient frames. Prevention: 1.Working set model (allocate based on locality), 2.Page fault frequency control, 3.Reduce multiprogramming.

Q12. File allocation methods. (5M)

- Contiguous: Sequential blocks, fast, external frag, fixed size. Linked: Scattered blocks linked, no frag, slow random access, pointer overhead. Indexed (i-node): Index block stores all pointers, random access, overhead for small files. UNIX uses i-node.

Q13. Critical section problem. Requirements? (3M)

- Critical section: Code accessing shared resource. Problem: Race condition. Requirements: 1. Mutual Exclusion: only one process at a time. 2. Progress: decision made in finite time. 3. Bounded Waiting: limit on times others can enter before requester.

Q14. Semaphore. Binary vs counting. (3M)

- Semaphore: Integer variable for synchronization. wait(s): s--; if s<0 block. signal(s): s++; if s<=0 wake. Binary: 0/1 (mutex). Counting: 0 to N (resource count). Solves: mutual exclusion, ordering, bounded buffer.

Q15. Producer-Consumer problem solution. (5M)

- 3 semaphores: mutex=1(mutual exclusion), full=0(filled slots), empty=N(empty slots). Producer: wait(empty);wait(mutex);add_item;signal(mutex);signal(full); Consumer: wait(full);wait(mutex);remove_item;signal(mutex);signal(empty); Ensures no overflow, underflow, race condition.

Q16. Dining Philosophers problem. (3M)

- 5 philosophers, 5 chopsticks. Need 2 adjacent. Deadlock if all pick left. Solutions: 1. Allow ≤ 4 simultaneously. 2. Pick both or none (atomic). 3. Asymmetric: odd picks left first, even picks right first. Illustrates deadlock + starvation.

Q17. Disk scheduling — SCAN vs C-SCAN. (3M)

- SCAN (Elevator): Head moves in one direction servicing requests, reaches end, reverses. Unfair to recently-passed requests. C-SCAN: Moves in one direction, jumps to other end without servicing, more uniform. C-LOOK: SCAN but reverses at last request (not end of disk).

Q18. Types of OS. (3M)

- Batch: Jobs submitted as batches, no user interaction. Time-sharing: Multiple users, CPU switching. Real-time: Strict timing (RTOS). Distributed: Multiple computers as one. Mobile: Android/iOS. Embedded: Specific hardware (routers, microcontrollers).

Q19. Monolithic vs Microkernel. (3M)

- Monolithic: All OS services in kernel space. Fast, complex, error = crash. Example: Linux. Microkernel: Only essential in kernel (IPC, memory, CPU scheduling), rest in user space. Stable, slow (user-kernel communication). Example: Mach, QNX.

Q20. Explain internal vs external fragmentation. (3M)

- Internal: Allocated memory > required. Wasted inside partition. Caused by fixed-size partitions. Solution: Variable partitions or smaller blocks. External: Free memory available but not contiguous, can't satisfy request. Caused by variable partitions. Solution: Compaction, Paging.

4. LAST DAY REVISION BLOCK

- Process states: New→Ready→Running→Waiting→Terminated | PCB: stores process info
- Scheduling: FCFS(convoy), SJF(starvation), RR(fair), Priority(starvation→aging)
- Deadlock: 4 Coffman conditions | Banker's: safe state check
- Paging: no external frag | Segmentation: logical, external frag
- Page fault → load from disk | Thrashing → too many page faults
- FIFO: Belady's | LRU: best practical | OPT: theoretical
- Semaphore: wait=P (decrement) | signal=V (increment)

CA-125: Web Design-II

1. DEEP CORE NOTES

HTML5 Semantics

- Semantic tags: <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>, <figure>, <figcaption>
- HTML5 new input types: email, url, number, range, date, time, color, search, tel
- HTML5 new elements: <canvas>, <video>, <audio>, <progress>, <meter>, <datalist>, <details>, <summary>
- Canvas: Bitmap drawing via JS. context=canvas.getContext('2d'); ctx.fillRect(x,y,w,h)
- SVG: Vector-based. Scalable, text-searchable, animatable. For logos, icons.

CSS3 Advanced

- Box model: margin | border | padding | content. box-sizing: border-box includes padding+border in width.
- Selectors: *, .class, #id, element, [attr], :hover, :nth-child(n), ::before, ::after
- Flexbox: display: flex. flex-direction, justify-content, align-items, flex-wrap, gap. Responsive layout.
- Grid: display: grid. grid-template-columns, grid-template-rows, grid-gap, grid-column, grid-row.
- Transitions: transition: property duration timing-function delay; Smooth state changes.
- Animations: @keyframes name{from{}to{}} animation: name duration timing iteration direction;
- Transform: translate(x,y), rotate(deg), scale(x,y), skew(x,y)
- Media queries: @media (max-width:768px){} — Responsive design breakpoints.
- CSS variables: :root{--color:red;} use: color:var(--color);

JavaScript Core

- var (function scope, hoisted), let (block scope), const (block scope, no reassign)
- == (loose equality, type coercion) vs === (strict equality, no coercion)
- Hoisting: var declarations and function declarations moved to top of scope.
- Closure: Function that retains access to its outer scope variables after outer fn returns.
- Arrow functions: (x)=>x*2 — no own this, concise syntax.
- Destructuring: const {a,b}=obj; const [x,y]=arr;
- Spread/Rest: ...arr (spread into args), ...args (rest params), {...obj} (clone)
- Promises: new Promise((resolve,reject)=>{}) .then(cb) .catch(cb) .finally(cb)
- async/await: async fn(){let r=await fetch(url); return r.json();} — syntactic sugar for promises.

DOM Manipulation

- document.getElementById, querySelector, querySelectorAll, getElementsByClassName
- element.innerHTML, textContent, style, classList (add/remove/toggle/contains)
- createElement, appendChild, removeChild, insertBefore, replaceChild
- addEventListener('event', callback). Event object: e.target, e.preventDefault(), e.stopPropagation()
- Event bubbling: child→parent. Capturing: parent→child. Delegation: attach to parent, check target.

Bootstrap / Responsive

- Bootstrap grid: 12-column. Breakpoints: xs(<576px), sm(≥576), md(≥768), lg(≥992), xl(≥1200), xxl(≥1400)
- Container: fixed-width. container-fluid: full-width. row→col structure.
- Utility classes: m-*, p-*, text-*, d-*, flex-*, bg-*, border-*
- Components: Navbar, Card, Modal, Form, Table, Button, Badge, Alert, Carousel.

jQuery

- \$(selector).method(). \$(document).ready(fn). Short: \$(fn).
- Selectors: \$('#id'), \$('.class'), \$('tag'), \$('[attr=val]')
- DOM: .html(), .text(), .val(), .attr(), .css(), .addClass(), .removeClass()
- Events: .click(), .keyup(), .on('event', fn), .hover(in,out)
- Ajax: \$.ajax({url,method,data,success,error}); \$.get(); \$.post(); \$.getJSON()
- Effects: .show(), .hide(), .toggle(), .fadeIn(), .fadeOut(), .slideUp(), .slideDown(), .animate()

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. HTML5 semantic elements. Why use? (3M)

- <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>. Benefits: Better SEO (search engines understand structure), Accessibility (screen readers navigate better), Cleaner code, Browser consistency. Replace generic <div id='header'> with meaningful tags.

Q2. Flexbox properties explained. (5M)

- Container: display:flex; flex-direction(row/column); justify-content(flex-start/center/space-between/around/evenly); align-items(stretch/center/flex-start/end); flex-wrap(wrap/nowrap); gap. Item: flex-grow, flex-shrink, flex-basis, align-self, order. One-dimensional layout.

Q3. CSS Grid vs Flexbox. When to use? (3M)

- Grid: 2D layout (rows + columns). Better for page layout. grid-template-columns: repeat(3,1fr). Flexbox: 1D (row or column). Better for component layout, nav bars, card alignment. Use both: Grid for outer layout, Flexbox for inner components.

Q4. CSS animation vs transition. (3M)

- Transition: Two states, triggered by event (hover). transition:all 0.3s ease; Smooth A→B. Animation: Multiple keyframes, auto-starts. @keyframes bounce{0%{top:0}50%{top:20px}100%{top:0}} animation:bounce 1s infinite; More control.

Q5. var vs let vs const. (3M)

- var: function scope, hoisted (initialized undefined), can re-declare. let: block scope, not hoisted (TDZ), no re-declare. const: block scope, must initialize, no reassign (but object properties can change). Use const by default, let when reassign needed, avoid var.

Q6. Explain closures with example. (5M)

```
function counter(){
  let count=0;
  return function(){
    count++;
    return count;
  };
}
const c=counter();
console.log(c()); // 1
console.log(c()); // 2
// Inner fn retains access to count even after counter() finished
```

- Uses: Data privacy, memoization, factory functions, event handlers with persistent state.

Q7. Event bubbling vs capturing. Delegation. (3M)

- Bubbling: Event propagates child→parent (default). Capturing: parent→child (3rd param true). stopPropagation() stops. Delegation: Attach one listener to parent, check event.target. Efficient for dynamic elements. Ex: document.querySelector('ul').addEventListener('click', e=>{ if(e.target.tagName==='LI') handle(e.target) })

Q8. Promises and async/await. (5M)

- Promise: Object for async operation. States: pending, fulfilled, rejected. Chain: .then().catch(). async/await: syntactic sugar. async fn(){try{const r=await fetch(url); const d=await r.json(); return d;}catch(e){console.error(e);}} More readable than promise chains.

Q9. Bootstrap grid system. (3M)

- 12-column grid. <div class='container'><div class='row'><div class='col-md-6'>. col-xs/sm/md/lg/xl. col-6 = 50%, col-4 = 33%. Responsive: col-md-4 col-sm-12. No column = auto-width. offset-md-3 = 3 col margin left.

Q10. jQuery Ajax example. (5M)

```
$.ajax({
  url: 'api/data',
  method: 'GET',
  dataType: 'json',
  success: function(data){
    $('#result').html(data.message);
  },
  error: function(xhr,status,err){
    console.log('Error:'+err);
  }
});
// OR shorthand:
$.getJSON('api/data', function(data){...});
```

Q11. DOM manipulation. Add element dynamically. (3M)

```
const li = document.createElement('li');
li.textContent = 'New Item';
li.classList.add('active');
document.querySelector('#list').appendChild(li);
// Remove: element.parentNode.removeChild(element)
// Or: element.remove()
```

Q12. HTML5 Canvas. Draw shapes. (3M)

```
<canvas id='c' width='400' height='200'></canvas>
<script>
```

```
const ctx = document.getElementById('c').getContext('2d');
ctx.fillStyle = 'blue';
ctx.fillRect(10,10,100,80); // filled rect
ctx.strokeStyle = 'red';
ctx.strokeRect(150,10,100,80); // outline rect
ctx.beginPath();
ctx.arc(300,50,40,0,Math.PI*2); // circle
ctx.fill();
</script>
```

Q13. Responsive design. Media queries. (3M)

- Viewport: <meta name='viewport' content='width=device-width,initial-scale=1'>. Media queries: @media (max-width:768px){.container{width:100%;padding:10px;}} Mobile-first: design for small, add complexity for larger. Breakpoints: 576, 768, 992, 1200px.

Q14. LocalStorage vs SessionStorage vs Cookies. (3M)

- localStorage: Persistent, 5-10MB, no expiry, same origin. sessionStorage: Per tab/session, 5MB, cleared on close. Cookies: 4KB, sent with HTTP request, has expiry, accessible server-side. Use: Auth→cookies(httpOnly), Preferences→localStorage, Cart(session)→sessionStorage.

Q15. CSS specificity. (3M)

- Specificity hierarchy: !important > inline style (1,0,0,0) > #id (0,1,0,0) > .class/:pseudo (0,0,1,0) > element (0,0,0,1). Higher specificity wins. Same specificity → last declared wins. Avoid !important; it breaks cascade.

Q16. What is AJAX? How it works? (3M)

- Asynchronous JavaScript And XML. Updates page without reload. Flow: JS creates XMLHttpRequest → sends HTTP request → server responds → JS handles response → updates DOM. fetch() API is modern alternative. Enables SPAs, live search, form validation.

Q17. JSON. Parse and stringify. (2M)

- JSON: JavaScript Object Notation. Data interchange format. JSON.parse(str) → object. JSON.stringify(obj) → string. Rules: Keys must be strings (double quotes), no functions/undefined/symbols. Used in APIs, localStorage, AJAX.

Q18. Prototype chain in JavaScript. (3M)

- Every JS object has [[Prototype]]. Property lookup: own → prototype → prototype's prototype → null. Object.create(proto). class syntax is syntactic sugar for prototype-based inheritance. hasOwnProperty() checks only own. For..in iterates inherited too.

Q19. Write jQuery form validation. (5M)

```
$('#form').on('submit', function(e){
    e.preventDefault();
    const name = $('#name').val().trim();
    const email = $('#email').val().trim();
    if(name===''){ alert('Name required'); return; }
    if(!/^[^@]+@^[^@]+\.[a-z]{2,}$/i.test(email)){
        alert('Invalid email'); return;
    }
    // process form
});
```

Q20. Explain event loop in JavaScript. (5M)

- JS is single-threaded. Event loop: Call Stack (running code) + Web APIs (setTimeout,fetch) + Callback Queue + Microtask Queue. Order: Call stack executes → empty → microtasks (Promise .then) → macrotasks (setTimeout). Explains why setTimeout(fn,0) runs after synchronous code.

3. PRACTICAL / PROGRAMS

P1: Responsive Navbar with Bootstrap

```
<nav class='navbar navbar-expand-lg navbar-dark bg-dark'>
  <a class='navbar-brand' href='#'>Logo</a>
  <button class='navbar-toggler' type='button' data-bs-toggle='collapse' data-bs-
target='#nav'>
    <span class='navbar-toggler-icon'></span>
  </button>
  <div class='collapse navbar-collapse' id='nav'>
    <ul class='navbar-nav ms-auto'>
      <li class='nav-item'><a class='nav-link' href='#'>Home</a></li>
    </ul>
  </div>
</nav>
```

P2: Flexbox Card Layout

```
.card-container{
  display:flex;
  flex-wrap:wrap;
  gap:20px;
  justify-content:center;
}
.card{
  width:250px;
  padding:20px;
  border:1px solid #ccc;
  border-radius:8px;
  transition:transform 0.3s;
}
.card:hover{transform:scale(1.05);}
```

P3: Fetch API JSON Display

```
async function loadData(){
  try{
    const res = await fetch('https://jsonplaceholder.typicode.com/posts/1');
    const data = await res.json();
    document.getElementById('output').textContent = data.title;
  }catch(e){
    console.error('Failed:', e);
  }
}
loadData();
```

4. LAST DAY REVISION BLOCK

- Semantic HTML5: header,nav,main,section,article,aside,footer
- Flexbox=1D (row/col) | Grid=2D | Bootstrap=12col grid
- var=function scope | let/const=block scope | const=no reassign
- Closure: inner fn retains outer scope | Hoisting: var+fn declarations moved up
- Promise states: pending/fulfilled/rejected | async/await = sugar for promises
- AJAX: async HTTP without reload | JSON.parse ↔ JSON.stringify
- Event: bubbling(up) | capturing(down) | delegation(parent listener)

CA-126: Introduction to Graphics Design

1. DEEP CORE NOTES

Fundamentals of Graphic Design

- Graphic Design: Visual communication using typography, images, color, space to convey message.
- Elements: Line, Shape, Color, Texture, Space (positive/negative), Typography, Image.
- Principles: Balance (symmetric/asymmetric), Contrast, Alignment, Repetition, Proximity, Hierarchy, White space.
- CRAP: Contrast, Repetition, Alignment, Proximity — Robin Williams' design principles.

Color Theory

- Primary: Red, Blue, Yellow (traditional). Digital RGB: Red, Green, Blue (additive).
- Secondary: Orange(R+Y), Green(B+Y), Purple(R+B). Tertiary: Primary+Secondary.
- Color wheel: Analogous (adjacent), Complementary (opposite), Triadic (equidistant 3), Split-complementary.
- Warm colors: Red, Orange, Yellow — energy, urgency. Cool: Blue, Green, Purple — calm, trust.
- Color models: RGB (screen), CMYK (print), HSB/HSL (hue-saturation-brightness), LAB.
- RGB: 0-255 each. White=255,255,255. Black=0,0,0. CMYK: 0-100%. K=Key(Black).

Typography

- Font vs Typeface: Typeface = design family (Helvetica). Font = specific weight/size (Helvetica Bold 12pt).
- Serif: Decorative strokes (Times New Roman) — formal, print. Sans-serif: No strokes (Arial) — modern, screen.
- Script: Cursive/handwritten. Monospace: Equal width (Courier). Display: Decorative/headline.
- Hierarchy: Headline→Subheading→Body. Contrast size, weight, color.
- Leading: Line spacing. Kerning: Space between two chars. Tracking: Space across word/sentence.

Image Types

Type	Based On	Scaling	Formats	Use
Raster/Bitmap	Pixels	Loses quality	JPEG,PNG,GIF,BMP	Photos
Vector	Math paths	Infinite scale	SVG,AI,EPS	Logos,icons

- Resolution: PPI (screen), DPI (print). Screen: 72/96 PPI. Print: 300 DPI minimum.
- Pixel: Smallest display unit. Megapixel = 1 million pixels.

Adobe Photoshop

- Raster editing. Tools: Selection (M), Lasso (L), Magic Wand (W), Crop (C), Brush (B), Clone Stamp (S), Healing Brush (J), Pen (P), Text (T), Gradient (G).
- Layers: Non-destructive editing. Layer types: Normal, Adjustment, Fill, Shape, Text, Smart Object.
- Blending modes: Normal, Multiply (darken), Screen (lighten), Overlay, Soft Light.
- Adjustments: Levels, Curves, Hue/Saturation, Color Balance, Brightness/Contrast, Photo Filter.
- Selection: Marquee, Lasso, Quick Select, Magic Wand, Select&Mask, Color Range, Pen path.
- Filters: Gaussian Blur, Unsharp Mask, Liquify, Smart Sharpen. Applied non-destructively via Smart Objects.
- Masks: Layer mask (grayscale), Vector mask (path), Clipping mask (clip to below layer).
- Shortcut keys: Ctrl+T (transform), Ctrl+J (duplicate layer), Ctrl+Z (undo), Ctrl+Shift+N (new layer), Alt+Delete (fill fg), Ctrl+Delete (fill bg).

Adobe Illustrator

- Vector editor. Tools: Selection (V), Direct Selection (A), Pen (P), Pencil (N), Text (T), Rectangle (M), Ellipse (L), Paintbrush (B).
- Anchor points: Corner vs Smooth. Bezier curves: direction handles control curve.
- Pathfinder: Unite, Minus Front, Intersect, Exclude, Divide — combine shapes.
- Artboard: Like Photoshop canvas. Multiple artboards for multiple pages/sizes.
- Symbols, Swatches, Brushes, Graphic Styles — Reusable assets.

File Formats

Format	Type	Key Feature
JPEG	Raster	Lossy, photos, no transparency
PNG	Raster	Lossless, transparency support
GIF	Raster	Animation, 256 colors
SVG	Vector	Scalable, web-friendly, editable
PSD	Raster	Photoshop native, layers

AI	Vector	Illustrator native
PDF	Both	Print/share, universal
TIFF	Raster	Lossless, high quality print
EPS	Vector	Legacy print format

Design Workflow

- Brief → Research → Sketch/Wireframe → Digital Draft → Feedback → Refine → Export → Deliver
- Mockup: Realistic representation of design on real-world object.
- Prototype: Interactive model showing user interaction flow.

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. What is graphic design? Elements? (3M)

- Visual communication discipline using text, images, color, space. Elements: Line (direction/movement), Shape (geometric/organic), Color (emotion/identity), Texture (surface feel), Space (positive=content, negative=empty), Typography, Image/Illustration.

Q2. Explain principles of design. (5M)

- Balance: Symmetric (formal) or Asymmetric (dynamic). Contrast: Difference creates interest. Alignment: Organize elements. Repetition/Consistency: Unify design. Proximity: Related elements grouped. Hierarchy: Guide eye through design. White space: Breathing room, reduces clutter.

Q3. RGB vs CMYK. When use each? (3M)

- RGB: Red, Green, Blue. Additive (light). Screen/digital. White=all colors. CMYK: Cyan, Magenta, Yellow, Key(Black). Subtractive (ink). Print use. White=no ink (paper). Design: Use RGB for web, convert to CMYK before print. RGB gamut > CMYK.

Q4. Raster vs vector graphics. (5M)

- Raster: Pixel grid. Fixed resolution. Scaling up = blur/pixelate. PPI matters. JPEG,PNG,BMP,PSD. Best for photos. Vector: Mathematical paths/equations. Resolution independent. Scale infinitely without quality loss. SVG,AI,EPS,PDF. Best for logos, icons, illustrations.

Q5. Explain color theory. Color harmonies. (5M)

- Color wheel: 12 colors (3P+3S+6T). Harmonies: Monochromatic=tints/shades of one hue, Analogous=adjacent (natural), Complementary=opposite (high contrast, vibrant), Triadic=3 equidistant (balanced+colorful), Split-complementary=safer alt to complementary, Tetradic=4 colors(rich).

Q6. Typography basics. Font types. (5M)

- Serif (Times): Formal, print-friendly. Sans-serif (Helvetica): Modern, digital. Script (Brush Script): Elegant, limited use. Monospace (Courier): Code, typewriter. Display: Headlines only. Hierarchy: Size+Weight+Color. Max 2-3 fonts per design. Pair: Sans+Serif often works.

Q7. Photoshop layers. Types? (3M)

- Layers: Non-destructive editing. Each element on separate layer. Types: Pixel layer (photos/drawing), Adjustment layer (Levels/Hue non-destructive), Fill layer (solid color/gradient/pattern), Shape layer (vector), Text layer, Smart Object (embedded/linked). Order matters: top layer on top.

Q8. Selection tools in Photoshop. (3M)

- Marquee: Rectangular/Elliptical geometric selection. Lasso: Freehand. Polygonal Lasso: Straight edges. Magnetic Lasso: Auto-snaps edge. Magic Wand: Color-based. Quick Select: Brush-based. Select & Mask: Refine hair/fur edges. Color Range: Select by color. Pen path: Most precise.

Q9. Blending modes in Photoshop. (3M)

- Normal: No blending. Multiply: Darkens (like overlapping transparencies). Screen: Lightens (like projected light). Overlay: Contrast+saturation boost. Soft Light: Gentle overlay. Hard Light: Strong. Luminosity: Uses top layer brightness. Color: Uses top layer color.

Q10. What is layer mask? (3M)

- Grayscale image that controls layer visibility. White=show, Black=hide, Gray=partial transparency. Non-destructive alternative to eraser. Applied by: Layer menu→Layer Mask→Reveal All/Hide All. Paint black to hide, white to reveal. Used for compositing, background removal.

Q11. Pen tool and Bezier curves. (3M)

- Pen tool: Creates precise paths using anchor points. Click=corner, Click+drag=smooth curve. Bezier curves controlled by direction handles. Paths can be: Stroked (outline), Filled (shape), Converted to selection. Used in Photoshop+Illustrator for precise cutouts and vector art.

Q12. Pathfinder in Illustrator. (3M)

- Shape Modes: Unite (combine), Minus Front (subtract top), Intersect (keep overlap), Exclude (remove overlap). Pathfinders: Divide (cut by all paths), Trim (remove hidden), Merge, Crop, Outline, Minus Back. Used to create complex shapes from simple ones.

Q13. What is DPI? Importance? (2M)

- DPI: Dots Per Inch. Measure of print resolution. Screen: 72-96 PPI. Web: 72 PPI. Print: 300 DPI minimum. Large format: 150 DPI. High DPI=sharper, larger file. Low DPI for print = pixelated/blurry output. Set at start of project.

Q14. Difference: JPEG vs PNG. (2M)

- JPEG: Lossy compression. No transparency. Best for photos. Smaller file. PNG: Lossless. Supports transparency (alpha channel). Larger file. Best for logos, icons, graphics with transparency. GIF: Animation, 256 colors only.

Q15. What is kerning, tracking, leading? (3M)

- Kerning: Spacing between specific letter pair (A-V needs reduction). Tracking: Uniform spacing across word/range. Leading: Vertical spacing between lines (line height). Poor leading = hard to read. Tight tracking: headings only. Normal: 120-145% leading of font size.

Q16. Smart Objects in Photoshop. (2M)

- Smart Object: Layer that preserves source content. Can scale/transform without losing quality. Apply filters non-destructively (Smart Filters). Linked smart objects update globally. Right-click→Convert to Smart Object. Cannot paint directly; must rasterize.

Q17. Design process steps. (3M)

- 1.Brief: Understand client need 2.Research: Audience, competitors, inspiration 3.Sketch: Rough ideas 4.Wireframe: Layout structure 5.Design: Digital execution 6.Feedback: Client review 7.Revise: Refine based on feedback 8.Export: Correct format/specs 9.Deliver: Final assets.

Q18. What is white space? Why important? (2M)

- White space (negative space): Empty area in design. Functions: Focus attention, improve readability, convey elegance/luxury, reduce cognitive load. Not wasted space; it's deliberate. Less = cluttered. More = professional, breathable design (Apple style).

Q19. Adjustment layers vs direct adjustments. (3M)

- Direct: Image→Adjustments→Curves. Destructive, permanent. Adjustment layer: Non-destructive, can edit/delete anytime, has built-in mask, affects all layers below. Always prefer adjustment layers. Types: Levels, Curves, Hue/Sat, Color Balance, Brightness, Vibrance, B&W.

Q20. Explain color psychology in design. (5M)

- Red: Energy, urgency, passion. Used by food brands (McDonald's), sales. Blue: Trust, calm, professional. Used by tech/banking (Facebook, IBM). Green: Nature, health, growth. Yellow: Optimism, warmth, warning. Black: Luxury, power. White: Clean, minimal. Purple: Creativity, royalty. Orange: Enthusiasm, affordable.

4. LAST DAY REVISION BLOCK

- Design elements: Line, Shape, Color, Texture, Space, Typography, Image
- CRAP: Contrast, Repetition, Alignment, Proximity
- RGB=screen(additive) | CMYK=print(subtractive) | RGB gamut > CMYK
- Raster=pixels(JPEG/PNG) | Vector=math paths(SVG/AI) — vector scales infinitely
- 300 DPI print | 72 PPI web | Layers=non-destructive editing
- Mask: White=show, Black=hide | Smart Object=non-destructive transform
- Kerning=pair spacing | Tracking=word spacing | Leading=line spacing

IC-125: भारतीय संविधान (Constitution of India - Marathi)

1. सखोल मुख्य नोट्स

संविधानाची पार्श्वभूमी

- संविधान सभा: 9 डिसेंबर 1946 रोजी पहिली बैठक. अध्यक्ष: डॉ. राजेंद्रप्रसाद.
- मसुदा समिती: डॉ. बाबासाहेब आंबेडकर (अध्यक्ष). बी.एन. राव: घटनातज्ज्ञ सल्लागार.
- संविधान निर्मिती: 2 वर्ष, 11 महिने, 18 दिवस. 26 नोव्हेंबर 1949 रोजी स्वीकृत (Law Day).
- अंमलबजावणी: 26 जानेवारी 1950 (Republic Day). मूळ: 395 कलमे, 22 भाग, 8 अनुसूची.
- सध्याची स्थिती: 448 कलमे, 25 भाग, 12 अनुसूची, 104 घटनादुरुस्त्या.

उद्देशिका (Preamble)

- आपण भारताचे लोक... भारत एक संप्रभु, समाजवादी, धर्मनिरपेक्ष, लोकशाही, प्रजासत्ताक राज्य...
- उद्देश: न्याय (सामाजिक, आर्थिक, राजकीय), स्वातंत्र्य, समता, बंधुत्व.
- 42वी दुरुस्ती 1976: समाजवादी, धर्मनिरपेक्ष, अखंडता जोडले.
- केशवानंद भारती खटला 1973: उद्देशिका संविधानाचा भाग आहे. Basic Structure Doctrine.

मूलभूत हक्क (भाग III, कलम 12-35)

कलम	मूलभूत हक्क
14-18	समतेचा अधिकार (Right to Equality)
19-22	स्वातंत्र्याचा अधिकार (Right to Freedom)
23-24	शोषणाविरुद्ध अधिकार
25-28	धर्मस्वातंत्र्याचा अधिकार
29-30	सांस्कृतिक व शैक्षणिक अधिकार
32	घटनात्मक उपायांचा अधिकार (Dr. Ambedkar: 'Soul of Constitution')

- 44वी दुरुस्ती 1978: मालमत्तेचा अधिकार मूलभूत हक्कातून काढला → कलम 300A (कायदेशीर अधिकार).
- 86वी दुरुस्ती 2002: कलम 21A — 6-14 वर्षे वयोगटासाठी मोफत व सक्तीचे शिक्षण.

राज्य धोरणाची मार्गदर्शक तत्वे (DPSP, भाग IV, कलम 36-51)

- न्यायालयात लागू होत नाहीत पण राज्यशासनावर बंधनकारक नैतिक. आयर्लंडच्या संविधानावरून घेतले.
- Socialistic: 38, 39, 39A, 41, 42, 43, 43A, 47. Gandhian: 40, 43, 43B, 46, 47, 48. Liberal: 44, 45, 48A, 49, 50, 51.
- कलम 44: समान नागरी संहिता (UCC). कलम 45: 6 वर्षांखालील बालसंगोपन (86व्याने 14 वर्षांपर्यंत). कलम 48A: पर्यावरण संरक्षण.

मूलभूत कर्तव्ये (भाग IVA, कलम 51A)

- 42वी दुरुस्ती 1976: 10 कर्तव्ये जोडली. 86वी दुरुस्ती 2002: 11वे कर्तव्य (शिक्षण). USSR च्या संविधानावरून प्रेरित.
- न्यायालयात लागू होत नाहीत. उदाहरणे: संविधान मानणे, राष्ट्रगीत/झेंड्याचा सन्मान, पर्यावरण रक्षण, वैज्ञानिक दृष्टिकोन बाळगणे.

केंद्र शासन

- राष्ट्रपती: भाग V, कलम 52. अप्रत्यक्ष निवड. 5 वर्षे. कार्यकारी प्रमुख. आणीबाणी: कलम 352, 356, 360.
- पंतप्रधान: कलम 74, 75. मंत्रिमंडळ प्रमुख. लोकसभेत बहुमत असणे आवश्यक. वास्तविक कार्यकारी.
- संसद: राज्यसभा (250) + लोकसभा (552). लोकसभा: 5 वर्षे, थेट निवडणूक. राज्यसभा: कायमस्वरूपी (1/3 दर 2 वर्षांनी निवृत्त).
- सर्वोच्च न्यायालय: कलम 124. CJI नियुक्ती: राष्ट्रपती. 65 वर्षे सेवानिवृत्ती.

राज्य शासन

- राज्यपाल: कलम 153. राष्ट्रपतींकडून नियुक्ती. 5 वर्षे. राज्याचा संविधानिक प्रमुख.
- मुख्यमंत्री: वास्तविक कार्यकारी. विधानसभेत बहुमत. राज्यपाल नियुक्त करतात.
- उच्च न्यायालय: कलम 214. न्यायाधीश: 62 वर्षे सेवानिवृत्ती.

महत्वाच्या घटनादुरुस्त्या

दुरुस्ती	वर्ष	महत्व
1st	1951	मालमत्ता, भाषण स्वातंत्र्य निर्बंध
42nd	1976	समाजवादी, धर्मनिरपेक्ष, मूलभूत कर्तव्ये

44th	1978	मालमत्ता मूलभूत हक्कातून काढले
52nd	1985	पक्षांतर बंदी (10वी अनुसूची)
61st	1988	मतदान वय 21→18
73rd/74th	1992	पंचायत राज/नगरपालिका (11वी/12वी अनुसूची)
86th	2002	कलम 21A: 6-14 वर्षे शिक्षण
101st	2016	GST लागू
103rd	2019	EWS आरक्षण (10%)

आणीबाणी तरतुदी

- कलम 352 (राष्ट्रीय आणीबाणी): युद्ध/बाह्य आक्रमण/सशस्त्र विद्रोह. 44 कलम 352 सुधारित. मूलभूत हक्क 19 निलंबित.
- कलम 356 (राष्ट्रपती राजवट): राज्यघटनात्मक यंत्रणा अपयशी. विधानसभा विसर्जन. राज्यपालांच्या शिफारशीवर.
- कलम 360 (आर्थिक आणीबाणी): भारताची वित्तीय स्थिरता धोक्यात. फक्त एकदाही घोषित नाही.

2. 20 अत्यंत महत्वाचे प्रश्न व उत्तरे

Q1. भारतीय संविधानाची प्रमुख वैशिष्ट्ये. (5M)

- 1. जगातील सर्वात मोठे लिखित संविधान. 2. कठोर + लवचिक (संयुक्त). 3. संसदीय शासन प्रणाली. 4. संघराज्य + एकात्मक वैशिष्ट्ये. 5. मूलभूत हक्क+DPSP+कर्तव्ये. 6. धर्मनिरपेक्ष राज्य. 7. स्वतंत्र न्यायपालिका. 8. प्रौढ मताधिकार.

Q2. उद्देशिका म्हणजे काय? 42व्या दुरुस्तीचे महत्त्व. (3M)

- उद्देशिका: संविधानाची प्रस्तावना. संविधानाचा आत्मा, हेतू, तत्वज्ञान. केशवानंद भारती (1973): संविधानाचा भाग. 42वी दुरुस्ती 1976: 'समाजवादी', 'धर्मनिरपेक्ष', 'अखंडता' शब्द जोडले. उद्दिष्टे: न्याय, स्वातंत्र्य, समता, बंधुत्व.

Q3. समतेचा अधिकार. कलम 14-18. (5M)

- 14: कायद्यापुढे समता व समान संरक्षण. 15: धर्म/वंश/जात/लिंग/जन्मस्थान भेदभाव निषेध. 16: सरकारी नोकरीत समान संधी. 17: अस्पृश्यता निर्मूलन (दंडनीय). 18: पदव्या रद्द (लष्करी/शैक्षणिक वगळता). 15(3)(4): महिला व मागास वर्गासाठी विशेष तरतूद.

Q4. स्वातंत्र्याचा अधिकार. कलम 19. (5M)

- कलम 19: 6 स्वातंत्र्ये: (a)भाषण/अभिव्यक्ती (b)शांततापूर्ण सभा (c)संघटना स्थापना (d)देशभर संचार (e)वास्तव्य (f)व्यवसाय. सर्वांवर वाजवी निर्बंध शक्य. 20: दुहेरी शिक्षा नाही. 21: जीव व स्वातंत्र्य (Right to Life). 22: अटक झाल्यावर हक्क.

Q5. घटनात्मक उपायांचा अधिकार (कलम 32). (3M)

- मूलभूत हक्कांचे रक्षण. डॉ. आंबेडकर: 'संविधानाचा आत्मा'. सर्वोच्च न्यायालयास 5 रिट जारी करण्याचा अधिकार: Habeas Corpus (बेकायदेशीर अटक), Mandamus (कर्तव्य बजावणे), Prohibition (अधिकारक्षेत्र), Certiorari (रद्द करणे), Quo-Warranto (पद वैधता).

Q6. DPSP म्हणजे काय? महत्त्व? (3M)

- राज्य धोरणाची मार्गदर्शक तत्वे. भाग IV, कलम 36-51. न्यायालयात लागू नाही पण शासनावर नैतिक बंधन. कल्याणकारी राज्याचे उद्दिष्ट. आयर्लंडवरून प्रेरित. UCC(44), UPE(45), Panchayati Raj(40), Workers Participation(43A) महत्वाची.

Q7. मूलभूत कर्तव्ये (51A). (3M)

- 42वी दुरुस्ती 1976: 10 कर्तव्ये. 86वी 2002: 11वे. USSR वरून प्रेरित. न्यायालयात लागू नाही. महत्वाची: संविधान/राष्ट्रगीत सन्मान, राष्ट्रीय चळवळ जपणे, अखंडता रक्षण, पर्यावरण, वैज्ञानिक दृष्टिकोन, 6-14 वर्षे मुलांना शिक्षण.

Q8. राष्ट्रपतींचे अधिकार व आणीबाणी. (5M)

- 352: राष्ट्रीय आणीबाणी (युद्ध/आक्रमण/विद्रोह). 356: राज्यघटनात्मक यंत्रणा अपयश = President's Rule. 360: आर्थिक आणीबाणी. आणीबाणी दरम्यान: संसद राज्यांसाठी कायदे करू शकते (State List), मूलभूत हक्क 19 निलंबित (352 दरम्यान). 44Cth: 352 कठोर = 2/3 बहुमत + लेखी मंत्रिमंडळ शिफारस.

Q9. पंचायत राज. 73वी दुरुस्ती. (5M)

- 73वी दुरुस्ती 1992: त्रिस्तरीय पंचायत राज संवैधानिक. ग्रामपंचायत→पंचायत समिती→जिल्हा परिषद. 11वी अनुसूची: 29 विषय. राज्य निवडणूक आयोग, 5 वर्षे कार्यकाळ, SC/ST/महिला आरक्षण (1/3 किमान). 74वी: नगरपालिका, 12वी अनुसूची: 18 विषय.

Q10. मूलभूत हक्क vs DPSP. (3M)

- मूलभूत हक्क: Part III, न्यायालयात लागू, व्यक्तिगत स्वातंत्र्य, नकारात्मक (राज्याला प्रतिबंध). DPSP: Part IV, न्यायालयात लागू नाही, सामाजिक/आर्थिक, सकारात्मक (राज्याने करायचे). संघर्ष: मिन्हा मिल्स खटला (1980): दोन्ही संतुलित असणे गरजेचे.

Q11. 42वी घटनादुरुस्ती. 'Mini Constitution' का? (3M)

- 1976 (Indira Gandhi). उद्देशिकेत 3 शब्द. मूलभूत कर्तव्ये. DPSP बळकट. संसदेला अमर्याद शक्ती. न्यायपालिका मर्यादित. Administrative Tribunals. मोठ्या बदलांमुळे Mini Constitution. पण 44वी दुरुस्ती 1978 ने अनेक बदल पूर्ववत.

Q12. लोकसभा vs राज्यसभा. (5M)

- लोकसभा: खालचे सभागृह. 543 निर्वाचित+2 नामनिर्देशित. थेट निवडणूक. 5 वर्षे. अर्थ विधेयक: लोकसभेत. सर्वशक्तिमान (no-confidence, appropriation). राज्यसभा: वरचे सभागृह. 238 राज्ये+12 राष्ट्रपती नामनिर्देशित. 6 वर्षे (1/3 दर 2 वर्षांनी). कायमस्वरूपी. विशेष अधिकार: 249, 312.

Q13. न्यायिक पुनर्विलोकन (Judicial Review). (3M)

- न्यायालयाचा कायदे/कार्यकारी कृती संविधानाशी सुसंगत आहेत का ते तपासण्याचा अधिकार. कलम 13, 32, 226, 131-136. केशवानंद भारती: Basic Structure असुरक्षित. USA वरून प्रेरित पण भारतात मर्यादित. संसदीय सर्वोच्चता vs न्यायिक पुनर्विलोकन संतुलन.

Q14. धर्मनिरपेक्षता भारतात. (3M)

- कलम 25-28: धर्म स्वातंत्र्य. राज्य कोणताही धर्म राजकीय धर्म म्हणून मान्य करत नाही. 42वी दुरुस्ती: 'Secular' उद्देशिकेत. कलम 15: धर्मावर भेदभाव नाही. कलम 44: UCC उद्दिष्ट. भारतीय धर्मनिरपेक्षता = Sarva Dharma Samabhav (सर्व धर्म समभाव), पाश्चात्य वेगळे.

Q15. SC/ST आरक्षण. संविधानिक तरतूद. (3M)

- कलम 15(4): शैक्षणिक संस्थांत. कलम 16(4): सरकारी नोकऱ्यांत. कलम 330/332: लोकसभा/विधानसभेत जागा. SC: 15%, ST: 7.5% (लोकसंख्येनुसार). OBC: कलम 16(4) मंडल आयोग 27%. EWS: 103वी दुरुस्ती 2019 → 10%. एकूण: 50% मर्यादा (Indra Sawhney Case).

Q16. मूलभूत रचना सिद्धांत (Basic Structure). (5M)

- केशवानंद भारती विरुद्ध केरळ राज्य (1973). 13 न्यायाधीशांचे पीठ. संसदेला संविधान दुरुस्त करण्याचा अधिकार आहे पण मूलभूत रचना बदलता येणार नाही. मूलभूत रचना: उद्देशिका, मूलभूत हक्क, धर्मनिरपेक्षता, लोकशाही, संघराज्य, न्यायिक पुनर्विलोकन, विभाजन of powers.

Q17. RTI कायदा 2005. (3M)

- Right to Information. 30 दिवसांत माहिती. BPL: 5 रुपये. कलम 19(1)(a) – भाषण स्वातंत्र्यातून. केंद्रीय माहिती आयोग/राज्य माहिती आयोग. 8 सूट: राष्ट्रीय सुरक्षा, संसद/न्यायालय अभिलेख, fiduciary, व्यक्तिगत माहिती. अपील: PIO→Appellate Authority→CIC.

Q18. प्रस्तावनेतील 'संप्रभु, समाजवादी, धर्मनिरपेक्ष, लोकशाही, प्रजासत्ताक'. (5M)

- संप्रभु: बाह्य/अंतर्गत सर्वोच्च सत्ता. समाजवादी: आर्थिक न्याय, संपत्तीचे समान वितरण. धर्मनिरपेक्ष: सर्व धर्म समान, राज्याचा धर्म नाही. लोकशाही: जनतेची सत्ता. प्रजासत्ताक: निवडून आलेला प्रमुख (राष्ट्रपती), वंशपरंपरागत नाही.

Q19. निवडणूक आयोग. (3M)

- कलम 324. स्वतंत्र घटनात्मक संस्था. मुख्य निवडणूक आयुक्त + 2 आयुक्त. 6 वर्षे कार्यकाळ किंवा 65 वर्षे. महाभियोग जसे CJI काढणे. अधिकार: मतदार यादी, आचारसंहिता, मान्यता, परिणाम घोषणा, पक्ष मान्यता/नाकारणे.

Q20. भारतीय राज्यघटनेची निर्मिती कशी झाली? (5M)

- 1946: संविधान सभा निर्मिती (Cabinet Mission Plan). डिसेंबर 1946: पहिली बैठक. Dr. Rajendra Prasad: अध्यक्ष. Dr. Ambedkar: मसुदा समिती अध्यक्ष. 2 वर्षे 11 महिने 18 दिवस. 11 सत्रे, 165 दिवस. 7635 सुधारणा, 2473 स्वीकृत. 26 Nov 1949: संमत. 26 Jan 1950: लागू.

4. अंतिम दिवस रिव्हिजन

- संविधान: 26 Nov 1949 (स्वीकृत) | 26 Jan 1950 (लागू) | Dr. Ambedkar: मसुदा समिती
- मूलभूत हक्क: Part III (12-35) | DPSP: Part IV (36-51) | कर्तव्ये: Part IVA (51A)
- DPSP: न्यायालयात लागू नाही | मूलभूत हक्क: लागू होतात
- 42वी: समाजवादी+धर्मनिरपेक्ष+कर्तव्ये | 44वी: मालमत्ता हक्क काढला
- 73वी: Panchayat | 74वी: Nagarpalika | 86वी: कलम 21A शिक्षण
- Basic Structure: केशवानंद भारती 1973 | RTI: 2005 | GST: 101वी 2016

CA-124: Basic Mathematics for Computing

1. DEEP CORE NOTES

Set Theory

- Set: Collection of distinct, well-defined objects. Notation: $A=\{1,2,3\}$ or $A=\{x:x\in Z, x>0\}$
- Types: Empty ($\emptyset$), Singleton, Finite, Infinite, Universal (U), Power set $P(A)=2^n$ subsets.
- Operations: Union ($A\cup B$), Intersection ($A\cap B$), Difference ($A-B=A\cap B'$), Complement (A').
- De Morgan's: $(A\cup B)'=A'\cap B'$, $(A\cap B)'=A'\cup B'$
- Inclusion-Exclusion: $|A\cup B|=|A|+|B|-|A\cap B|$. Three sets: $|A\cup B\cup C|=|A|+|B|+|C|-|A\cap B|-|B\cap C|-|A\cap C|+|A\cap B\cap C|$

Relations & Functions

- Relation: $R\subseteq A\times B$. Types: Reflexive (aRa), Symmetric ($aRb\rightarrow bRa$), Transitive ($aRb, bRc\rightarrow aRc$), Antisymmetric.
- Equivalence relation: Reflexive + Symmetric + Transitive.
- Partial order: Reflexive + Antisymmetric + Transitive (POSET).
- Function: Each element of domain maps to exactly one element of codomain.
- Types: Injective (1-1, $f(a)=f(b)\rightarrow a=b$), Surjective (onto, every y has preimage), Bijective (both).
- Composition: $(g\circ f)(x)=g(f(x))$. Inverse: f^{-1} exists iff f is bijective.

Mathematical Logic

- Proposition: Statement that is T or F. Propositional variables: p, q, r.
- Connectives: $\neg$ (NOT), $\wedge$ (AND), $\vee$ (OR), $\rightarrow$ (Implication), $\leftrightarrow$ (Biconditional)

p	q	$p\wedge q$	$p\vee q$	$p\rightarrow q$	$p\leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

- $p\rightarrow q \equiv \neg p\vee q$. Contrapositive: $\neg q\rightarrow\neg p \equiv p\rightarrow q$. Converse: $q\rightarrow p$. Inverse: $\neg p\rightarrow\neg q$.
- Tautology: Always T. Contradiction: Always F. Contingency: Sometimes T, sometimes F.
- Equivalences: $p\wedge T\equiv p$, $p\vee F\equiv p$, $p\wedge\neg p\equiv F$, $p\vee\neg p\equiv T$, $p\rightarrow q\equiv\neg p\vee q$

Predicate Logic

- Predicate: $P(x)$ — property of x. Quantifiers: $\forall x$ (Universal, 'for all'), $\exists x$ (Existential, 'there exists')
- $\neg\forall x P(x) \equiv \exists x \neg P(x)$. $\neg\exists x P(x) \equiv \forall x \neg P(x)$ — Quantifier negation.

Number Theory

- Divisibility: $a|b$ if $b=ka$ for some integer k. Properties: $a|b, b|c \rightarrow a|c$
- Prime: Divisible only by 1 and itself. Composite: Has other factors.
- GCD: $\gcd(a,b)$. Euclidean algo: $\gcd(a,b)=\gcd(b, a \bmod b)$. LCM= $a*b/\gcd(a,b)$
- Modular arithmetic: $a\equiv b \pmod n$ iff $n|(a-b)$. $(a+b) \bmod n = ((a \bmod n) + (b \bmod n)) \bmod n$
- Fermat's Little Theorem: If p prime and $\gcd(a,p)=1$, then $a^{p-1}\equiv 1 \pmod p$

Combinatorics

- Multiplication rule: $k_1 \times k_2 \times \dots \times k_n$ for sequential choices.
- Permutation: $P(n,r)=n!/(n-r)!$. Ordered selection r from n.
- Combination: $C(n,r)=n!/(r!(n-r)!)$. Unordered selection.
- Binomial theorem: $(x+y)^n=\sum C(n,k) x^k y^{n-k}$, $k=0$ to n .
- Pigeonhole principle: $n+1$ objects in n holes $\rightarrow$ at least one hole has ≥ 2 .

Recurrence Relations

- Recurrence: Defines sequence by previous terms. $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ (2nd order linear).
- Fibonacci: $F(n)=F(n-1)+F(n-2)$, $F(0)=0$, $F(1)=1$. $\rightarrow 0, 1, 1, 2, 3, 5, 8, 13, \dots$
- Characteristic equation: $a_n=c_1 a_{n-1}+c_2 a_{n-2} \rightarrow r^2=c_1 r+c_2 \rightarrow$ solve for roots.

Graph Theory

- Graph $G=(V,E)$. V=vertices, E=edges. $|V|=n$, $|E|=m$.
- Degree: $\deg(v)$ =edges incident. Sum of degrees= $2|E|$ (Handshaking lemma).
- Types: Undirected, Directed (digraph), Weighted, Complete (K_n), Bipartite, Planar.
- Paths: Walk (repeat allowed), Trail (edges no repeat), Path (vertices no repeat).
- Euler path: Visits every edge exactly once. Exists if 0 or 2 vertices have odd degree.
- Hamiltonian path: Visits every vertex exactly once. NP-complete to find.

- Tree: Connected graph with no cycles. n vertices $\rightarrow n-1$ edges. Spanning tree: subgraph.
- BFS/DFS: Graph traversal algorithms. BFS uses queue (level-order). DFS uses stack/recursion.
- Planar graph: Drawn without edge crossing. Euler's formula: $V-E+F=2$ (connected planar).

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. Prove De Morgan's law for sets. (3M)

- $(A \cup B)' = A' \cap B'$. Proof: $x \in (A \cup B)' \leftrightarrow x \notin A \cup B \leftrightarrow x \notin A \wedge x \notin B \leftrightarrow x \in A' \wedge x \in B' \leftrightarrow x \in A' \cap B'$. Hence proved. Similarly $(A \cap B)' = A' \cup B'$.

Q2. Inclusion-Exclusion principle. Example. (3M)

- $|A \cup B| = |A| + |B| - |A \cap B|$. Example: 100 students, 60 Math, 50 Sci, 30 both. Math or Sci = $60+50-30=80$. Neither = $100-80=20$.

Q3. Types of functions. Example each. (5M)

- Injective (1-1): $f(x)=2x$. No two inputs give same output. Surjective (onto): $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x)=x+1$. Every integer reached. Bijective: $f(x)=x+2$. Both injective+surjective. Only bijective functions have inverse. f^{-1} exists iff f is bijection.

Q4. Equivalence relation. Example. (3M)

- Relation R on A is equivalence if: Reflexive (aRa), Symmetric ($aRb \rightarrow bRa$), Transitive ($aRb, bRc \rightarrow aRc$). Example: 'same remainder mod n '. $a \equiv b \pmod{3}$: Reflexive: $a-a=0$ divisible by 3 $\checkmark$. Symmetric: $a-b$ divisible $\rightarrow b-a$ divisible $\checkmark$. Transitive: $a \equiv b, b \equiv c \rightarrow a \equiv c$ $\checkmark$.

Q5. Truth table for $p \rightarrow q$ and equivalences. (3M)

- $p \rightarrow q$ is F only when $p=T, q=F$. $p \rightarrow q \equiv \neg p \vee q$. Contrapositive $\neg q \rightarrow \neg p \equiv p \rightarrow q$ (logically equivalent). Converse $q \rightarrow p$ NOT equivalent. Inverse $\neg p \rightarrow \neg q$ NOT equivalent. Biconditional $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

Q6. What is tautology? Give example. (2M)

- Tautology: Compound proposition that is TRUE for all combinations of truth values. Example: $p \vee \neg p$ is always T (Law of excluded middle). $p \rightarrow p$ always T. $(p \wedge q) \rightarrow p$ always T. Contradiction: always F ($p \wedge \neg p$).

Q7. Euclidean algorithm. Find GCD(252, 105). (3M)

- $\gcd(a,b) = \gcd(b, a \bmod b)$. $\gcd(252, 105)$: $252 = 2 \times 105 + 42 \rightarrow \gcd(105, 42)$. $105 = 2 \times 42 + 21 \rightarrow \gcd(42, 21)$. $42 = 2 \times 21 + 0 \rightarrow \gcd = 21$. $\text{LCM}(252, 105) = 252 \times 105 / 21 = 1260$.

Q8. Permutation vs Combination. Examples. (3M)

- Permutation: Order matters. $P(n,r) = n!/(n-r)!$ $P(5,2) = 5!/3! = 20$. Combination: Order doesn't matter. $C(n,r) = n!/(r!(n-r)!)$ $C(5,2) = 10$. Committee of 3 from 8 = $C(8,3) = 56$. Arrange 3 people from 8 = $P(8,3) = 336$.

Q9. Pigeonhole principle. Example. (3M)

- $n+1$ objects in n containers $\rightarrow$ at least one has ≥ 2 . Generalized: $\lceil (n+1)/k \rceil$ in some container if $nk+1$ objects in n . Example: 13 people $\rightarrow$ 2 born in same month. 27 English words $\rightarrow$ 2 start with same letter.

Q10. Fibonacci recurrence. Solve $F(n)$. (3M)

- $F(n) = F(n-1) + F(n-2)$, $F(0)=0$, $F(1)=1$. Sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55... Characteristic equation: $r^2 = r + 1 \rightarrow r = (1 \pm \sqrt{5})/2$. Golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618$. $F(n) = (\phi^n - (1-\phi)^n)/\sqrt{5}$ (Binet's formula).

Q11. Euler path and Euler circuit conditions. (3M)

- Euler path (traverses every edge once): Exists if graph connected AND exactly 0 or 2 vertices have odd degree. Euler circuit (starts and ends same): Exists if connected AND ALL vertices have even degree. Example: Königsberg bridge problem — all 4 nodes odd degree $\rightarrow$ no Euler path.

Q12. Tree properties. (3M)

- Tree: Connected, acyclic graph. n vertices, $n-1$ edges. Unique path between any two vertices. Adding any edge creates a cycle. Removing any edge disconnects it. Spanning tree: Subgraph that includes all vertices, is a tree. A connected graph has at least one spanning tree.

Q13. Handshaking lemma. (2M)

- Sum of degrees of all vertices = $2|E|$. $\sum \deg(v) = 2m$. Consequence: Number of odd-degree vertices is always even. Proof: Each edge contributes 2 to total degree sum.

Q14. Complete graph K_n . Properties. (2M)

- K_n : Every pair of vertices connected. $n(n-1)/2$ edges. Each vertex degree = $n-1$. K_3 = triangle. K_4 = 4 vertices, 6 edges. K_5 is not planar. $K_{3,3}$ (complete bipartite) also not planar (Kuratowski).

Q15. Modular arithmetic. Find $7^{100} \bmod 5$. (3M)

- $7 \equiv 2 \pmod{5}$. $2^1=2$, $2^2=4$, $2^3=3$, $2^4=1 \pmod{5} \rightarrow$ period 4. $100 = 4 \times 25$, so $2^{100} \equiv (2^4)^{25} \equiv 1^{25} \equiv 1 \pmod{5}$. Therefore $7^{100} \equiv 1 \pmod{5}$. Fermat's Little: $a^{p-1} \equiv 1 \pmod{p}$. $p=5$, $7^4 \equiv 1 \pmod{5}$.

Q16. Bipartite graph. How to check? (3M)

- Bipartite: $V = V_1 \cup V_2$ where all edges between V_1 and V_2 , none within. 2-colorable. Theorem: G is bipartite iff it has no odd-length cycles. Check: BFS 2-coloring — color neighbor opposite color. Conflict = not bipartite. K -bipartite ($K_{m,n}$): $m \times n$ edges.

Q17. Planar graph. Euler's formula. (3M)

- Planar: Can draw without edge crossings. Euler's formula: $V-E+F=2$ (connected planar). V =vertices, E =edges, F =faces(including outer). K_4 : $4-4+3$? No. K_4 : $V=4, E=6, F=4$. $4-6+4=2$ ✓. Non-planar: K_5 (5 vertices, 10 edges, would need $F=7$, but $3F \leq 2E$). Kuratowski: contains K_5 or $K_{3,3}$ subdivision.

Q18. Power set. How many subsets of {1,2,3,4}? (2M)

- $P(A)$ = set of all subsets. $|P(A)|=2^n$. $\{1,2,3,4\}$: $2^4=16$ subsets. Includes $\emptyset$ and A itself. Power set $P(\{a,b\})=\{\emptyset, \{a\}, \{b\}, \{a,b\}\}$.

Q19. Solve recurrence $a_n=5a_{n-1}-6a_{n-2}$. (5M)

- Characteristic equation: $r^2=5r-6 \rightarrow r^2-5r+6=0 \rightarrow (r-2)(r-3)=0 \rightarrow r=2,3$. General solution: $a_n=A \times 2^n + B \times 3^n$. With initial conditions $a_0=1, a_1=5$: $A+B=1, 2A+3B=5 \rightarrow A=-2, B=3$. $a_n=-2 \times 2^n + 3 \times 3^n$.

Q20. BFS vs DFS for graph traversal. (5M)

- BFS: Uses queue. Level by level. Finds shortest path in unweighted graph. Time $O(V+E)$. Space $O(V)$. Applications: Shortest path, level-order, bipartite check. DFS: Uses stack/recursion. Depth first. Time $O(V+E)$. Space $O(V)$. Applications: Cycle detection, topological sort, SCC, maze solving.

4. LAST DAY REVISION BLOCK

- De Morgan: $(A \cup B)' = A' \cap B'$ | Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$
- $p \rightarrow q \equiv \neg p \vee q$ | Tautology=always T | Contradiction=always F
- $P(n,r) = n!/(n-r)!$ | $C(n,r) = n!/(r!(n-r)!)$ | Pigeonhole: $n+1$ in n
- GCD: Euclidean algo | $LCM = a \cdot b / GCD$
- Tree: n vertices, $n-1$ edges, connected, acyclic
- Euler path: 0 or 2 odd-degree | Euler circuit: all even degree
- Planar: $V-E+F=2$ | Handshaking: $\sum \deg = 2|E|$

ENG-123: English for All - Written Communication

1. DEEP CORE NOTES

Written Communication Basics

- Written communication: Permanent record. Precise, planned, formal. Needs clarity, coherence, correctness.
- 7 Cs: Clear, Concise, Concrete, Correct, Complete, Coherent, Courteous.

Business Letters

- Parts: Letterhead, Date, Inside address, Salutation, Subject, Body (intro/body/closing), Complimentary close, Signature.
- Formats: Full Block (all left), Modified Block (date/close right), Semi-Block (indent paragraphs).
- Types: Inquiry, Order, Complaint, Adjustment, Sales, Application, Resignation, Recommendation.
- Salutation: Dear Mr./Ms./Dr. [Surname]. Close: Yours sincerely (named), Yours faithfully (unnamed).

Report Writing

- Report: Formal document with facts, findings, analysis, recommendations.
- Structure: Title, Date, To/From, Summary/Abstract, Introduction, Methodology, Findings, Analysis, Conclusions, Recommendations, Appendices.
- Types: Informational (facts only), Analytical (with analysis), Annual, Progress, Incident, Technical, Feasibility.
- Formal report: Lengthy, third person, structured. Informal: Memo-like, shorter.

Email Writing

- Subject line: Clear, specific, concise. Body: Greeting → Purpose → Details → Action required → Closing.
- Professional tone. Avoid: ALL CAPS, slang, emoticons. CC: copy. BCC: blind copy. Reply-all: use carefully.
- Signature: Name, Designation, Company, Contact.

Essay Writing

- Structure: Introduction (hook + thesis), Body paragraphs (topic sentence + evidence + analysis), Conclusion (restate + implication).
- Types: Descriptive, Narrative, Expository, Argumentative, Persuasive.
- Paragraph structure: Topic sentence → Supporting sentences → Concluding sentence.
- Coherence: Use transitions — Furthermore, Moreover, However, In contrast, Therefore, Consequently, In conclusion.

Précis Writing

- Précis: Compressed version of passage. 1/3 original length. Own words. Third person, past tense.
- Steps: Read carefully → identify theme → note key points → write in own words → check length → give title.
- Don'ts: No personal opinions, no direct quotes, no examples from original, no figurative language.

Grammar - Key Rules

- Tenses: Simple, Continuous (ing), Perfect (have/had), Perfect Continuous. 12 tenses in 4×3 matrix.
- Active/Passive: Active: Subject+verb+object. Passive: Object+be+past participle. By+agent (optional).
- Direct/Indirect speech: Reporting verb change. Tense backshift. Pronouns change. Time expressions change (now→then, today→that day, tomorrow→next day).
- Articles: a (consonant sound), an (vowel sound), the (specific/unique). No article: proper nouns, abstract nouns (generally).
- Subject-verb agreement: Singular subject=singular verb. Collective nouns=singular. Either/Neither=singular. Each/Every=singular.

Notice, Agenda, Minutes

- Notice: Formal announcement. Parts: Organization name, Notice (heading), Date, Title, Body, Name/Designation.
- Agenda: List of items to be discussed at meeting. Action Agenda includes decision column.
- Minutes: Official record of meeting. Types: Narrative, Action, Verbatim. Include: Attendance, quorum, decisions, next meeting.

2. 20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. What are the 7 Cs of communication? (3M)

- Clear: One interpretation. Concise: No unnecessary words. Concrete: Specific facts. Correct: Grammar/spelling/facts. Complete: All needed info. Coherent: Logical flow. Courteous: Respectful, polite. These principles ensure effective written communication.

Q2. Parts of a business letter. (5M)

- 1.Letterhead (company info), 2.Date, 3.Reference number, 4.Inside address (recipient), 5.Salutation (Dear Sir/Ms.), 6.Subject line, 7.Body (3 paras: purpose/details/action), 8.Complimentary close (Yours sincerely), 9.Signature, 10.Enclosures (if any). Full Block format: all left-aligned.

Q3. Write a letter of complaint. (8M)

- Model: Date: [Date]. To: Customer Service Manager. Subject: Complaint regarding defective laptop. Dear Sir/Madam, I purchased [product] on [date] vide invoice no. [no.]. I regret to inform you that the product has [defect]. This has caused inconvenience as [impact]. I request you to kindly [action: replace/refund/repair] at the earliest. I am enclosing a copy of the purchase invoice. I look forward to a prompt resolution. Yours faithfully, [Name].

Q4. Types of reports. (3M)

- By purpose: Informational (facts only), Analytical (interpretation). By time: Annual, Periodic, Progress. By subject: Technical, Financial, Marketing. By length: Long formal, Short informal (memo). By authority: Statutory (legally required), Non-statutory.

Q5. Structure of a formal report. (5M)

- Cover page → Title page → Letter of transmittal → Table of contents → List of tables/figures → Abstract/Executive summary → Introduction (background,scope,objectives) → Methodology → Findings/Results → Analysis/Discussion → Conclusions → Recommendations → References → Appendices.

Q6. What is précis? Rules for writing. (5M)

- Précis: Compressed summary in own words. 1/3 original length. Rules: Use own words, third person, past tense for reporting verbs, no personal opinion, no direct quotes, no examples, no figurative language, give a title, logical flow. Steps: Read → Theme → Key points → Draft → Check length → Title.

Q7. Professional email writing. Structure. (3M)

- Subject: Specific, actionable. Greeting: Dear [Name]/Hi [Name] (semi-formal). Opening line: Purpose. Body: Concise paragraphs. Action required: Clear CTA. Closing: Thank you, Regards/Best regards. Signature: Full contact. Avoid: excessive formatting, vague subject, no-reply tone.

Q8. Active to Passive voice. Rules and examples. (5M)

- Rule: Object→Subject + be(tense)+V3 + by+original subject. Ram writes a letter → A letter is written by Ram. She has eaten the cake → The cake has been eaten by her. They will complete the project → The project will be completed by them. Note: 'by' phrase optional if agent not important.

Q9. Direct to Indirect speech. Rules. (5M)

- Reporting verb change: says→says/said. Tense backshift: is→was, will→would, can→could, has→had. Pronouns: I→he/she, we→they, you→he/she. Time: now→then, today→that day, tomorrow→next day, yesterday→previous day. Questions: if/whether for yes/no, wh-word for wh-questions. No inverted commas.

Q10. Essay types and structure. (5M)

- Descriptive: Sensory details, paint picture. Narrative: Story with events. Expository: Explain/inform, balanced. Argumentative: Take a stand with evidence. Persuasive: Change reader's mind. Structure: Hook → Background → Thesis (intro) → Body (3 paras: claim+evidence+analysis) → Restated thesis + Broader implication (conclusion).

Q11. What is notice writing? Write a sample. (5M)

- [Organization Name]. NOTICE. Date: [Date]. [Title: e.g., Annual Function Postponed]. This is to inform all students that the Annual Function scheduled for [date] has been postponed to [new date] due to [reason]. All students are requested to note the change. [Name], [Designation]. Box the notice if exam requires.

Q12. Minutes of meeting. (3M)

- Official record. Contains: Meeting name, Date/Time/Venue, Attendees, Apologies, Quorum confirmation, Agenda items discussed (decision + action owner + deadline for each), Any Other Business, Next meeting date, Signature of chairperson/secretary. Action minutes most common in business.

Q13. Subject-verb agreement rules. (3M)

- Singular subjects → singular verb. Collective nouns (team, committee) → singular. Either/Neither → singular. Each/Every/Nobody/Everyone → singular. Compound subject (A and B) → plural. Amounts/distances as unit → singular. There is/are: verb agrees with following noun.

Q14. Use of articles (a, an, the). (3M)

- a: before consonant sound (a cat, a university). an: before vowel sound (an apple, an hour). the: specific noun (the book on table), unique (the sun), superlatives (the best). No article: proper nouns (India), plural countable general (Dogs are loyal), uncountable general (Water is life).

Q15. Transition words for essays. (2M)

- Addition: Furthermore, Moreover, In addition, Besides. Contrast: However, Nevertheless, On the other hand, Although. Cause/Effect: Therefore, Consequently, As a result, Hence. Time: Subsequently, Meanwhile, Previously. Conclusion: In conclusion, To summarize, Overall.

Q16. Types of business letters. (3M)

- Inquiry: Request for info/quotation. Order: Placing order. Complaint: Reporting problem. Adjustment/Claim: Response to complaint. Sales: Promote product. Application: Job application. Circular: Mass announcement. Credit/Collection: Payment matters. Recommendation: Endorsing someone.

Q17. CV/Resume writing tips. (3M)

- Sections: Personal info, Career objective, Education, Work experience, Skills, Projects, Certifications, Extracurricular, References. Use action verbs: Developed, Managed, Led, Created. Quantify: 'Increased sales by 20%'. Keep 1-2 pages. ATS-friendly: no tables, standard fonts. Tailor for each job.

Q18. Write a job application letter. (8M)

- Date. To: HR Manager, [Company]. Subject: Application for [Position]. Dear Sir/Madam, I am writing in response to the job posting for [position] on [source]. I have completed [degree] from [institution] with [percentage]. My key skills include [skill1, skill2, skill3]. I have [project/experience] where I [achievement]. I am confident that my [quality] will contribute to [company]. I have enclosed my CV for your consideration. I would welcome an opportunity for an interview. Yours sincerely, [Name].

Q19. What is a memorandum (memo)? (3M)

- Memo: Internal written communication within organization. Format: TO: / FROM: / DATE: / SUBJECT: / Body / Signature. No salutation, no complimentary close. Direct, brief, factual. For: policy changes, reminders, internal announcements, brief reports. Not as formal as letter.

Q20. Tenses — quick reference. (5M)

- Simple Present: I work. Present Continuous: I am working. Present Perfect: I have worked. Present Perfect Continuous: I have been working. Simple Past: I worked. Past Continuous: I was working. Past Perfect: I had worked. Simple Future: I will work. Future Perfect: I will have worked. Key: BE verb for continuous; HAVE/HAD for perfect.

4. LAST DAY REVISION BLOCK

- 7 Cs: Clear, Concise, Concrete, Correct, Complete, Coherent, Courteous
- Letter: named→Yours sincerely | unnamed→Yours faithfully
- Précis: 1/3 length, own words, third person, no opinion, no quotes
- Passive: Object + be(tense) + V3 + by + agent
- Indirect: tense backshift + pronoun change + time expression change
- Essay: Hook→Thesis (intro) | Claim+Evidence+Analysis (body) | Restate+Implication (conclusion)
- Articles: a=consonant sound | an=vowel sound | the=specific | no article=general

MAR-124: स्पर्धा परीक्षांसाठी व्याकरण भाग-II

1. सखोल मुख्य नोट्स

समास (Compound Words)

- समास: दोन किंवा अधिक शब्दांचा मिळून बनलेला एक नवा शब्द. विग्रह = समासाचे विभाजन.

समास प्रकार	वैशिष्ट्य	उदाहरण
अव्ययीभाव	पहिला शब्द अव्यय, अर्थ प्रमुख	यथाशक्ती, प्रतिदिन
तत्पुरुष	दुसरा शब्द प्रमुख	राजमार्ग, देशभक्ती
कर्मधारय	विशेषण + नाम, दोन्ही प्रमुख	नीलकमल, महापुरुष
द्विगु	संख्यावाचक + नाम	त्रिभुवन, पंचपाळे
द्वंद्व	दोन्ही शब्द समान महत्व	आईवडील, सुखदुःख
बहुव्रीही	अन्य शब्द प्रमुख (विशेषण)	चतुर्भुज, पद्मनयन

अलंकार (Figures of Speech)

अलंकार	व्याख्या	उदाहरण
उपमा	दोन वस्तूंची तुलना, साम्यवाचक शब्द	तो सिंहासारखा लढला
रूपक	तुलना पण साम्यवाचक शब्द नाही, एकरूपता	आयुष्य एक रणांगण आहे
उत्प्रेक्षा	एका गोष्टीत दुसरीची कल्पना, जणू/मानो	नयन जणू कमळाचीच फुले
अतिशयोक्ती	अतिरंजित वर्णन	त्याने डोंगर उचलला
विरोधाभास	विरुद्ध अर्थाचे शब्द	मुक्या जखमा बोलतात
यमक	समान उच्चारचे वेगळे अर्थ	जरी जरी रेशीम जरी
श्लेष	एकाच शब्दाचे दोन अर्थ	हरि (देव/माकड) आला
अनुप्रास	एकाच वर्णाची पुनरावृत्ती	चंद्र चमके चांदणे चारूनयने

वाक्यप्रचार व म्हणी (Idioms & Proverbs)

- वाक्यप्रचार उदाहरणे: अंगाशी येणे (नुकसान होणे), डोळ्यांत तेल घालणे (दक्ष राहणे), हात टेकणे (हार मानणे), कानावर हात ठेवणे (नकळणे), पोटात ढवळणे (संताप होणे).
- म्हणी उदाहरणे: नाव सोनुबाई हाती कथलाचा वाळा (बडेजाव), एका हाताने टाळी वाजत नाही, गाढवापुढे वाचली गीता, कळ सोसता फळ मिळे, दुधाने तोंड भाजले म्हणून ताकही फुंकून प्यावे.

काळ (Tenses)

काळ	उदाहरण
वर्तमानकाळ	मी जातो / मी जात आहे / मी गेलो आहे
भूतकाळ	मी गेलो / मी जात होतो / मी गेलो होतो
भविष्यकाळ	मी जाईन / मी जात जाईन / मी गेलो असेन

वाक्यांचे प्रकार

- केवल वाक्य: एकच उद्देश + विधेय. राम शाळेत गेला.
- संयुक्त वाक्य: दोन स्वतंत्र वाक्ये जोडलेली. मी आलो आणि त्याने पाहिले.
- मिश्र/गुंफित वाक्य: एक मुख्य + एक/अधिक गौण वाक्ये. जो मेहनत करतो तो यशस्वी होतो.

भाषांतर / Translation

- मराठी→इंग्रजी: व्याकरण नियम पाळणे. काळ, कारक, समास लक्षात ठेवणे.
- शब्दशः नाही; भावानुवाद करणे. म्हणींचे समकक्ष इंग्रजी शोधणे.

शब्दसंपत्ती

- विरुद्धार्थी शब्द: सत्य↔असत्य, प्रकाश↔अंधार, ज्ञान↔अज्ञान, श्रेष्ठ↔कनिष्ठ, सुख↔दुःख.
- समानार्थी शब्द: सूर्य=रवि=दिनकर=भास्कर | पाणी=जल=नीर=तोय | आकाश=गगन=अंबर=नभ
- अनेकवचन: मुलगा→मुले, पुस्तक→पुस्तके, गाव→गावे, माणूस→माणसे, स्त्री→स्त्रिया

उभयान्वयी अव्यय (Conjunctions)

- समुच्चयबोधक: आणि, व, तसेच, शिवाय. विकल्पबोधक: किंवा, नाहीतर. न्यूनत्वबोधक: पण, परंतु. कारणबोधक: कारण, म्हणून. परिणामबोधक: तेव्हा, म्हणजे. उद्देशबोधक: की, यासाठी, जेणेकरून.

प्रत्यय व उपसर्ग

- उपसर्ग (Prefix): अ-/अन- (नाकारण): अज्ञान, अनादर. सु- (चांगले): सुमन. दुर्- (वाईट): दुर्गुण. प्र-: प्रगती. परि-: परिसर.
- प्रत्यय (Suffix): -पणा: शहाणपणा. -त्व: मानवत्व. -ता: स्वतंत्रता. -कार: कवीकार. -वान: बुद्धिमान.

2. 20 अत्यंत महत्वाचे प्रश्न व उत्तरे

Q1. समास म्हणजे काय? तत्पुरुष समास स्पष्ट करा. (3M)

- समास = दोन शब्दांचे अर्थपूर्ण एकत्रीकरण. तत्पुरुष: दुसरा शब्द प्रमुख. विग्रहात कारकविभक्ती असते. उदा: राजमार्ग (राजाचा मार्ग - षष्ठी तत्पुरुष), देशभक्ती (देशावरील भक्ती), सभागृह (सभेचे गृह).

Q2. कर्मधारय व बहुव्रीही समासातील फरक. (3M)

- कर्मधारय: दोन्ही पद एकाच वस्तूसाठी. विशेषण+नाम. नीलकमल=नीळे कमळ (कमळच). बहुव्रीही: दोन्ही पद अन्य वस्तूसाठी विशेषण. चतुर्भुज=चार भुजा ज्याला तो (विष्णू). 'तो/ती/ते' ने ओळखा.

Q3. उपमा व रूपक अलंकारातील फरक. (3M)

- उपमा: तुलना + साम्यवाचक शब्द (सारखा, जसा, प्रमाणे). 'तो सिंहासारखा लढला'. रूपक: साम्यवाचक शब्द नाही. एकरूपता. 'जीवन एक युद्ध आहे'. उपमेय = उपमान म्हणून सादर. 'तो सिंहच आहे'.

Q4. अतिशयोक्ती अलंकार. (2M)

- वास्तवाहून अतिरंजित वर्णन करणे. 'त्याने डोंगर एकट्याने उचलला', 'रक्ताचे पूर वाहिले', 'त्याच्या पराक्रमाने दिशा थरथरल्या'. उद्देश: भावना तीव्र करणे.

Q5. द्वंद्व समास सोदाहरण. (2M)

- दोन्ही पद समान महत्वाचे. दोन्ही अर्थ विग्रहात येतात. उदाहरणे: आईवडील (आई आणि वडील), सुखदुःख (सुख आणि दुःख), आगपाणी, रात्रंदिवस, उन्हाळा-हिवाळा, देश-विदेश.

Q6. खालील वाक्यप्रचारांचे अर्थ व वाक्य. (5M)

- हात टेकणे: हार मानणे. 'शेवटी तो हात टेकला.' डोळ्यांत तेल घालणे: दक्षता ठेवणे. 'पोलीस डोळ्यांत तेल घालून पाहला देतात.' अंगाशी येणे: नुकसान होणे. 'अति धाडसाचे अंगाशी येते.' पोटात ढवळणे: तीव्र संताप. 'ते दृश्य पाहून पोटात ढवळले.' कानावर हात ठेवणे: नकळण्याचे नाटक. 'त्याने कानावर हात ठेवले.'

Q7. पुढील म्हणींचे अर्थ सांगा. (5M)

- नाव सोनुबाई हाती कथलाचा वाळा: बाह्य थाट, आतून रिकामे. एका हाताने टाळी वाजत नाही: एकट्याने भांडण होत नाही. कळ सोसता फळ मिळे: कठोर परिश्रमाशिवाय यश नाही. गाढवापुढे वाचली गीता: निरुपयोगी माणसास उपदेश. दुधाने तोंड भाजले म्हणून ताकही फुंकून प्यावे: वाईट अनुभवाने अति सावधगिरी.

Q8. भूतकाळाचे प्रकार उदाहरणांसह. (3M)

- साधा भूत: मी गेलो. अपूर्ण भूत: मी जात होतो. पूर्ण भूत: मी गेलो होतो. आनुमानिक भूत: मी गेलो असेन. हेतुहेतुमदभूत: जर मी गेलो असतो तर... रीती: तो रोज जात असे.

Q9. केवल, संयुक्त, मिश्र वाक्य. (3M)

- केवल वाक्य: एकच विधान. 'राम खेळतो.' संयुक्त वाक्य: 'राम खेळतो आणि श्याम झोपतो.' मिश्र वाक्य: 'जो अभ्यास करतो तो पास होतो.' — गौण+मुख्य. मिश्रातील गौण वाक्याची ओळख: जो...तो, जेव्हा...तेव्हा, जर...तर, की.

Q10. विरुद्धार्थी शब्द द्या. (3M)

- सत्य↔असत्य | ज्ञान↔अज्ञान | प्रकाश↔अंधार | उपकार↔अपकार | श्रेष्ठ↔कनिष्ठ | यश↔अपयश | आशा↔निराशा | जन्म↔मृत्यू | सुंदर↔कुरूप | आदर↔अनादर | शांती↔अशांती | सुरक्षित↔असुरक्षित

Q11. समानार्थी शब्द द्या. (3M)

- सूर्य: रवि, दिनकर, भास्कर, मार्तंड, तपन, अंशुमाली. पाणी: जल, नीर, उदक, तोय, अंबु. आकाश: गगन, नभ, अंबर, व्योम, आभाळ. पृथ्वी: धरा, वसुंधरा, भूमी, धरित्री. झाड: वृक्ष, तरू, पादप, द्रुम.

Q12. यमक व श्लेष अलंकार. (3M)

- यमक: समान उच्चारचे वेगळ्या अर्थाचे शब्द. 'जरी जरी रेशीम जरी' — जरी(कपडा/जरी-तरी/जरी=सुवर्ण). श्लेष: एकच शब्द दोन वेगळ्या अर्थानी. 'हरि आला' — हरि=देव किंवा माकड. काव्यात दोन्ही अर्थ सयुक्तिक.

Q13. उपसर्ग वापरून शब्द बनवा. (3M)

- अ/अन: अज्ञान, अनादर, अहंकार, अनुभव. सु: सुमन, सुलभ, सुविधा, सुरक्षित. दुर्/दुस्: दुर्गुण, दुस्तर, दुर्लभ, दुर्गम. प्र: प्रगती, प्रभाव, प्रयत्न, प्रतिमा. परि: परिस्थिती, परिसर, परिणाम, परिश्रम. वि: विकास, विज्ञान, विश्वास, विनंती.

Q14. वाक्यरूपांतर: भूत → भविष्य. (2M)

- राम शाळेत गेला. → राम शाळेत जाईल. मी जेवण केले. → मी जेवण करेन. त्यांनी निर्णय घेतला. → ते निर्णय घेतील. वर्तमान: गेला→जातो→जाईल. 'होते/होता' → 'आहे/असतो' → 'असेल/राहील'

Q15. वाक्यांतरः कर्तरी → कर्मणी. (3M)

- कर्तरी वाक्यः कर्त्याला प्रमुखता. मी पत्र लिहिले. कर्मणीः कर्माला प्रमुखता. माझ्याकडून पत्र लिहिले गेले. नियमः कर्त्याला 'कडून', कर्म = नवा कर्ता, क्रियापद + 'जाणे/गेले'. राम आंब्याला खातो → रामाकडून आंब्याला खाल्ले जाते.

Q16. अव्ययीभाव समास. (2M)

- पहिला शब्द अव्यय (प्र, यथा, प्रति, अनु, आ, सहित). अर्थावर भर. उदाहरणे: यथाशक्ती (शक्तीप्रमाणे), प्रतिदिन (दिवसादिवशी), आजन्म (जन्मापासून), भरपेट (पोट भरेपर्यंत), निःसंदेह. क्रियाविशेषण अव्यय म्हणून वापर.

Q17. संधी म्हणजे काय? उदाहरणे. (3M)

- संधी: दोन स्वर/व्यंजन लागून त्यांचे बदल होणे. स्वर संधी: विद्या+आलय=विद्यालय, देव+इंद्र=देवेन्द्र. व्यंजन संधी: जगत्+नाथ=जगन्नाथ. विसर्ग संधी: तेजः+मय=तेजोमय. मराठी संधी: 'न' चा 'ण': रामण (राम+न).

Q18. शब्दयोगी अव्यय (Postpositions). (2M)

- नामानंतर येणारे अव्यय. मुळे: कारण. करता/साठी: उद्देश. शिवाय: विरहित. पर्यंत: मर्यादा. बदल: विषयी. सारखा: साम्य. सहित/समवेत: बरोबर. विरुद्ध: प्रतिकूल. उदा: त्याच्यामुळे, त्यासाठी, माझ्याशिवाय, घरापर्यंत.

Q19. अनुप्रास अलंकार. (2M)

- एकाच व्यंजनाची पुनरावृत्ती. 'पाहे पान पाखरू पुळके' — 'प' ची पुनरावृत्ती. 'चंद्र चमके चांदणे चारू' — 'च' ची. काव्यात लयबद्धता आणते. ध्वनिसौंदर्य हा उद्देश. प्रकार: वृत्तानुप्रास (एकाच वर्णाची), छेकानुप्रास, लाटानुप्रास.

Q20. परिच्छेद लेखन: 'पाण्याचे महत्व'. (8M)

- पाणी हे जीवन आहे. पृथ्वीवरील सर्व सजीवांना पाण्याशिवाय जगणे अशक्य आहे. मानवी शरीरात ७०% पाणी असते. पिण्यासाठी, शेतीसाठी, उद्योगासाठी पाण्याचे अनन्यसाधारण महत्व आहे. आज जगात पाण्याची टंचाई ही गंभीर समस्या आहे. भारतातील अनेक भागांत दुष्काळ पडतो. पाण्याची बचत करणे, पाणी प्रदूषण टाळणे, पावसाचे पाणी साठवणे या उपायांनी आपण भविष्यातील पाणी टंचाई टाळू शकतो. तेव्हा प्रत्येकाने पाण्याचा काटकसरीने वापर करावा.

4. अंतिम दिवस रिव्हिजन

- समास: अव्ययीभाव, तत्पुरुष, कर्मधारय, द्विगु, द्वंद्व, बहुव्रीही
- उपमा: साम्यवाचक शब्द आहे | रूपक: नाही | उत्प्रेक्षा: जणू/मानो
- बहुव्रीही ओळख: 'तो/ती/ते' ने विग्रह | कर्मधारय: दोन्ही एकाच वस्तूसाठी
- कर्तरी: कर्त्याला महत्व | कर्मणी: 'कडून', क्रियापद+जाणे
- केवल=एक वाक्य | संयुक्त=आणि/पण | मिश्र=जो...तो/जर...तर
- उपसर्ग: अ/अन, सु, दुर्, प्र, परि, वि | प्रत्यय: -पणा, -त्व, -ता, -वान